

A polymer gas approach to $\mathbb{Z}_N$ homological quantum codes

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Abstract

We study finite-temperature P -form $\mathbb{Z}_N$ homological codes via an exact finite-Trotter quantum-to-classical map to a $(d + 1)$ -dimensional spacetime model with electric and magnetic topological background charges. We show that these background-dependent partition functions admit an exact reformulation as a gas of closed magnetic and electric defect polymers that interact via linking phases. By bounding this complex polymer gas by positive same-species hard-core majorants, we derive an explicit low-activity criterion under which all background-dependent partition functions are uniformly controlled and homologically nontrivial polymers are exponentially suppressed on the scale of the spacetime systole. We also derive an exact higher-form Kramers–Wannier duality exchanging electric and magnetic backgrounds, Wilson and ’t Hooft operators, and P -form and $(d - P)$ -form theories. Finally, for prime N , we identify an exact source-free gauge specialization coupled to the plaquette random-cluster model; on special geometries this connection brings sharp phase-transition results into the spacetime framework.

Contents

1	Introduction	2
2	Quantum model and decorated quantum-to-classical map	3
2.1	Code Hamiltonian with generic sources	4
2.2	Spatial Wilson and ’t Hooft operators	5
2.3	Trotter weights and neutral spacetime model	5
2.4	Spatial background	6
2.5	Euclidean twists and generic background	7
3	Polymer gas in topological backgrounds	9
3.1	Code limit and the flat vacuum	9
3.2	Local defect expansion	10
3.3	Closed defect expansion and homology constraints	11
3.4	Connected polymer expansion	12
3.5	Fourier resolution of the homology constraints	14
4	A rigorous low-activity region	15
4.1	Sectorwise gas and absolute majorization	15
4.2	Activity and counting bounds	16
4.3	The low-activity criterion	17
4.4	Large polymer suppression	20
5	Kramers-Wannier duality	21
5.1	Duality in the fixed-charge basis	21
5.2	Specialization to the Trotter weights	24
5.3	Duality of the quantum code	26

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6	Gauge/PRCM specializations	27
6.1	The exact gauge/PRCM specialization	27
6.2	Middle-dimensional toric specialization	29
7	Conclusion and outlook	32
A	Fourier conventions on $\mathbb{Z}_N$	32
B	Lattice geometry, duality, and linking	33
B.1	Chains, cochains, and the spacetime complex	33
B.2	Chain-cochain duality	35
B.3	Intersection pairing	37
B.4	Linking pairing	37
C	Supporting computations	40
C.1	Proof of Prop. 2.1	40
C.2	Untwisted Trotter kernels	41
C.3	Spatial Wilson and 't Hooft insertions	43
C.4	Euclidean twists	43
C.5	Fully decorated trace	44

1 Introduction

Homological quantum codes are quantum many-body systems whose ground-state degeneracy is organized by topology. Physical degrees of freedom are assigned to cells of a lattice, stabilizers are constructed from cellular boundary and coboundary operators, and logical operators are represented by nontrivial homology classes [1–5]. At zero temperature this gives a topological encoding of quantum memory. At finite temperature, however, the same topological structure becomes intertwined with statistical mechanics: thermal excitations form extended defects, and defect histories that are homologically nontrivial can alter the encoded data. This interplay between topological quantum memory, lattice gauge theory, duality, and defect expansions has long been central to the study of finite-temperature stability [3, 6–18].

In this paper we develop an exact finite-temperature spacetime framework for P -form $\mathbb{Z}_N$ homological codes on a finite cell decomposition Λ of a closed oriented d -manifold. We allow general local X - and Z -type source terms and study the thermal ensemble itself, rather than the effective disordered statistical-mechanical models used in decoding-threshold analyses and quenched disorder averages [3, 8, 19]. Our central objects are background-resolved spacetime partition functions obtained from the thermal trace by inserting Wilson and 't Hooft operators together with Euclidean twists around the thermal circle. At fixed Trotter number M , this decorated trace admits an exact quantum-to-classical reformulation (Prop. 2.4) as a classical model on the spacetime lattice

$$\underline{\Lambda} = \Lambda \times (\mathbb{Z}/M\mathbb{Z}), \quad (1.1)$$

with electric and magnetic spacetime backgrounds determined by the chosen insertions. The finite-temperature problem is therefore organized not by a single partition function but by a family of background-dependent spacetime partition functions indexed by spacetime homology classes, simultaneously encoding ordinary spatial logical operators and twists winding around Euclidean time. This part of the construction holds for arbitrary $N \geq 2$.

The main structural development of the paper is an exact reformulation of these spacetime partition functions in terms of extended defects. Expanding the local weights around the flat-field regime produces local magnetic and electric defect variables, and summing over the spacetime field forces them to assemble into closed defects whose total homology classes are fixed by the chosen electric and magnetic backgrounds. This closed-defect expansion can be expressed in terms of a generalized electric-magnetic linking pairing, and for prime N , decomposing the defects into support-connected components yields an exact two-species polymer gas of connected closed electric and magnetic polymers (Prop. 3.2, Prop. 3.3). This polymer description is the main framework developed here. It resolves the finite-temperature ensemble into spacetime topological sectors, rewrites thermal fluctuations in terms of closed extended objects, and provides a common language for perturbative analysis and exact

duality. In this picture, the thermally dangerous excitations are exactly the homologically nontrivial connected polymers.

A first application is perturbative control near the flat-field, or code, limit (Thm. 4.3). The exact two-species polymer gas is generally complex: opposite-species polymers interact through pure linking phases rather than positive Boltzmann weights. We control it by comparing its absolute value to a product of two positive same-species hard-core polymer gases and applying the criteria of Kotecký-Preiss, Fernández-Procacci, and Bissacot-Fernández-Procacci [20–23]. This yields an explicit low-activity region in which all background-dependent partition functions are uniformly controlled. In that regime, large connected polymers are exponentially suppressed, and homologically nontrivial polymers are suppressed on the scale of the corresponding spacetime systole (Cor. 4.5). The weakest point in the resulting region boundary is a purely combinatorial counting problem for rooted connected closed polymers, so improvements in that counting problem would directly sharpen the rigorous low-activity region.

A second general output is an exact higher-form Kramers-Wannier duality (Prop. 5.1, Thm. 5.2). The derivation is again exact and local: one Fourier-transforms the cell weights and reinterprets the Fourier variables as fields on the dual spacetime lattice. The result exchanges the P -form theory on $\underline{\Lambda}$ with a $(d - P)$ -form theory on the dual lattice $\underline{\Lambda}^\vee$, exchanges electric and magnetic backgrounds, and at the level of the quantum code interchanges X - and Z -type stabilizers, local Weyl operators, and Wilson and 't Hooft logical operators. On self-dual lattices this becomes an exact $\mathbb{Z}_2$ -symmetry of the full spacetime theory-space. This places the construction in the higher-form Kramers-Wannier/Wegner duality tradition [6, 7, 24–31]. Unlike the linking-based polymer reformulation, this duality holds for arbitrary $N \geq 2$.

The general spacetime framework is broader than a positive lattice model: it allows arbitrary local weights, complex phases, and fixed electric and magnetic backgrounds. It nevertheless contains natural exact positive slices. The most important is obtained by removing the P -cell weights and retaining flatness weights on $(P + 1)$ -cells, which yields P -form N -state Potts lattice gauge theory [6, 32, 33]. For prime N , this gauge theory is exactly coupled to the plaquette random-cluster model by a higher-form Fortuin-Kasteleyn/Edwards-Sokal construction [34–38]. This gauge/PRCM slice provides a sharp application of the general framework. In particular, for the middle-dimensional self-dual toric geometry $\underline{\Lambda} = T_L^{2(P+1)}$, Duncan-Schweinhart proved for odd prime N that the associated $(P + 1)$ -dimensional PRCM undergoes a sharp homological phase transition at the self-dual probability $p_{\text{sd}}(N) = \frac{\sqrt{N}}{1+\sqrt{N}}$ [37]. Through the exact gauge/PRCM coupling, this gives a corresponding benchmark on the gauge-theory slice and, in particular, a gauge-theoretic interpretation of the critical coupling in terms of macroscopic toric sectors and value-locking of suitably separated toric observables (Prop. 6.2).

Taken together, these results provide an exact finite-temperature sector decomposition for homological codes and related lattice models. The thermal problem is reorganized in terms of background-resolved spacetime partition functions, closed defect gases, polymer expansions, and exact duality. We expect this framework to be useful beyond the specific gauge/PRCM application developed here, both for sharper perturbative questions and for further nonperturbative specializations.

The paper is organized as follows. In §2 we define the quantum model, the Wilson/'t Hooft and Euclidean twist insertions, and the exact decorated quantum-to-classical map. In §3 we derive the local defect expansion, the closed-defect gas, and the connected polymer representation. In §4 we identify the low-activity region and prove the exponential suppression of large and homologically nontrivial polymers. In §5 we derive the exact higher-form Kramers-Wannier duality and its action on the quantum code. Finally, §6 identifies the exact gauge/PRCM specialization and develops the middle-dimensional toric application. The appendices collect the finite Fourier conventions, lattice geometry, generalized linking pairing, and Trotter-kernel computations used in the main text.

2 Quantum model and decorated quantum-to-classical map

In this section we pass from the quantum $\mathbb{Z}_N$ homological code on the spatial lattice Λ to an exact finite- M spacetime model on

$$\underline{\Lambda} = \Lambda \times S_M^1 \quad \text{where,} \quad S_M^1 := \mathbb{Z}/M\mathbb{Z}. \quad (2.1)$$

We define the quantum model, the spatial Wilson and 't Hooft operators, the Euclidean twist operators, and the resulting spacetime background partition functions.

Throughout, Λ is a finite cell decomposition of a closed oriented d -manifold, and the code degrees of freedom live on the P -cells of Λ . For a cell complex X , we write $C_p(X)$ for its oriented p -cells, $\mathcal{C}_p(X)$ for its p -chains, and $\Omega^p(X)$ for its p -cochains, as reviewed in Appendix B.1. All computations of this section, culminating in Prop. 2.4, work for any integer $N \geq 2$.

2.1 Code Hamiltonian with generic sources

To each P -cell $b \in C_P(\Lambda)$ we assign an N -dimensional Hilbert space $\mathcal{H}_b \cong \mathbb{C}^N$, and define

$$\mathcal{H} := \bigotimes_{b \in C_P(\Lambda)} \mathcal{H}_b. \quad (2.2)$$

Fix an orthonormal basis $\{|j\rangle\}_{j \in \mathbb{Z}_N}$ of $\mathbb{C}^N$, and set

$$\omega := e^{2\pi i/N}. \quad (2.3)$$

On $\mathbb{C}^N$, define the shift and clock operators by

$$\hat{X}|j\rangle = |j+1\rangle, \quad \hat{Z}|j\rangle = \omega^j|j\rangle, \quad j \in \mathbb{Z}_N. \quad (2.4)$$

They obey

$$\hat{X}^r \hat{X}^s = \omega^{rs} \hat{X}^s \hat{X}^r, \quad \hat{X}^N = \hat{Z}^N = \hat{1}, \quad \hat{X}^\dagger = \hat{X}^{-1}, \quad \hat{Z}^\dagger = \hat{Z}^{-1}. \quad (2.5)$$

For any operator $\hat{O}$ on $\mathbb{C}^N$, we write $\hat{O}_b$ for the operator on $\mathcal{H}$ acting as $\hat{O}$ on $\mathcal{H}_b$ and as the identity on all other tensor factors. We also define the Kronecker delta and the fundamental character of $\mathbb{Z}_N$:

$$\delta_N(x) := \begin{cases} 1, & x = 0 \in \mathbb{Z}_N, \\ 0, & x \neq 0 \in \mathbb{Z}_N, \end{cases} \quad \chi(x) := \omega^x. \quad (2.6)$$

For $a \in C_{P-1}(\Lambda)$ and $c \in C_{P+1}(\Lambda)$, define the stabilizer operators

$$\hat{A}_a := \prod_{b \supset a} \hat{X}_b^{(-1)^P \varepsilon(b,a)}, \quad \hat{B}_c := \prod_{b \subset c} \hat{Z}_b^{\varepsilon(c,b)}. \quad (2.7)$$

The sign $(-1)^P$ in $\hat{A}_a$ is the convention compatible with the vertical component of the spacetime coboundary D on $\underline{\Lambda} = \Lambda \times S_M^1$ (see (B.18)). The incidence identity $\partial^2 = 0$ implies that all $\hat{A}_a$ commute with all $\hat{B}_c$, and the two families are separately commuting.

Define the stabilizer projectors

$$\hat{\mathcal{A}}_a := \frac{1}{N} \sum_{m \in \mathbb{Z}_N} \hat{A}_a^m, \quad \hat{\mathcal{B}}_c := \frac{1}{N} \sum_{m \in \mathbb{Z}_N} \hat{B}_c^m. \quad (2.8)$$

Let $J_a > 0$ and $K_c > 0$ be couplings on $(P-1)$ - and $(P+1)$ -cells, and let $g_b^{(n)}, h_b^{(n)}$ be local X - and Z -source coefficients on P -cells. We define

$$\hat{H}_0 = - \sum_{a \in C_{P-1}(\Lambda)} J_a \hat{\mathcal{A}}_a - \sum_{c \in C_{P+1}(\Lambda)} K_c \hat{\mathcal{B}}_c, \quad (2.9)$$

$$\hat{H}_1 = - \sum_{b \in C_P(\Lambda)} \sum_{n \in \mathbb{Z}_N} \left(g_b^{(n)} \hat{X}_b^n + h_b^{(n)} \hat{Z}_b^n \right), \quad (2.10)$$

and the total Hamiltonian

$$\hat{H} := \hat{H}_0 + \hat{H}_1. \quad (2.11)$$

Another separation of the Hamiltonian is into its X - and Z -dependent parts:

$$\hat{H} = \hat{H}_x + \hat{H}_z, \quad (2.12)$$

with

$$\hat{H}_x = - \sum_{a \in C_{P-1}(\Lambda)} J_a \hat{\mathcal{A}}_a - \sum_{b \in C_P(\Lambda)} \sum_{n \in \mathbb{Z}_N} g_b^{(n)} \hat{X}_b^n, \quad (2.13)$$

and

$$\hat{H}_z = - \sum_{c \in C_{P+1}(\Lambda)} K_c \hat{B}_c - \sum_{b \in C_P(\Lambda)} \sum_{n \in \mathbb{Z}_N} h_b^{(n)} \hat{Z}_b^n. \quad (2.14)$$

All terms within $\hat{H}_x$ commute with one another, and all terms within $\hat{H}_z$ commute with one another. The formulas below make sense for complex source parameters. If one wants $\hat{H}$ Hermitian, one should impose

$$g_b^{(n)} = \overline{g_b^{(-n)}}, \quad h_b^{(n)} = \overline{h_b^{(-n)}}. \quad (2.15)$$

2.2 Spatial Wilson and 't Hooft operators

In the absence of sources, i.e., with $g = h = 0$, the code subspace is

$$\mathcal{H}_{\text{code}} := \bigcap_{a \in C_{P-1}(\Lambda)} \ker(\hat{A}_a - \hat{1}) \cap \bigcap_{c \in C_{P+1}(\Lambda)} \ker(\hat{B}_c - \hat{1}). \quad (2.16)$$

Let $\nu = \sum_{b \in C_P(\Lambda)} \nu_b b \in Z_P(\Lambda)$ be a spatial P -cycle and $\mu^\vee \in Z_{d-P}(\Lambda^\vee)$ be a $(d-P)$ -cycle of the dual lattice. Then Wilson and 't Hooft operators supported on ν and $\mu^\vee$ respectively can be defined as¹

$$\hat{\mathcal{W}}_\nu := \prod_{b \in C_P(\Lambda)} \hat{Z}_b^{\nu_b}, \quad \hat{\mathcal{T}}_{\mu^\vee} := \prod_{b \in C_P(\Lambda)} \hat{X}_b^{(-1)^P I_\Lambda(\mu^\vee, b)}. \quad (2.17)$$

Some well-known algebraic properties of these operators are

Proposition 2.1. *Let $\nu \in Z_P(\Lambda)$ and $\mu^\vee \in Z_{d-P}(\Lambda^\vee)$.*

- (i) *The operators $\hat{\mathcal{W}}_\nu$ and $\hat{\mathcal{T}}_{\mu^\vee}$ preserve the code subspace $\mathcal{H}_{\text{code}}$.*
- (ii) *If $\nu \in B_P(\Lambda)$, then $\hat{\mathcal{W}}_\nu$ is a product of $\hat{B}_c$ -stabilizers and hence acts trivially on $\mathcal{H}_{\text{code}}$. Therefore $\hat{\mathcal{W}}_\nu$ depends only on $[\nu] \in H_P(\Lambda)$ when restricted to the code subspace.*
- (iii) *If $\mu^\vee \in B_{d-P}(\Lambda^\vee)$, then $\hat{\mathcal{T}}_{\mu^\vee}$ is a product of $\hat{A}_a$ -stabilizers and hence acts trivially on $\mathcal{H}_{\text{code}}$. Therefore $\hat{\mathcal{T}}_{\mu^\vee}$ depends only on $[\mu^\vee] \in H_{d-P}(\Lambda^\vee)$ when restricted to the code subspace.*
- (iv) *The mixed commutator is*

$$\hat{\mathcal{T}}_{\mu^\vee} \hat{\mathcal{W}}_\nu = \omega^{-(-1)^P I_\Lambda(\mu^\vee, \nu)} \hat{\mathcal{W}}_\nu \hat{\mathcal{T}}_{\mu^\vee}. \quad (2.18)$$

These follow from elementary algebra, we reproduce the proof in Appendix C.1 for completion.

2.3 Trotter weights and neutral spacetime model

The thermal partition function at inverse temperature β is

$$Z := \text{tr}_{\mathcal{H}} \left(e^{-\beta \hat{H}} \right). \quad (2.19)$$

For $M \geq 1$, define the Trotterized partition function

$$Z_M := \text{tr}_{\mathcal{H}} \left[\left(e^{-\beta \hat{H}_z / M} e^{-\beta \hat{H}_x / M} \right)^M \right], \quad Z = \lim_{M \rightarrow \infty} Z_M. \quad (2.20)$$

The limit equation is an application of the Lie/Trotter product formula. All identities below are exact at fixed M .

We define local weight functions which we shall use to write traces such as (2.20). For horizontal $(P+1)$ -cells $c(i) = c \times \{i\}$ and horizontal P -cells $b(i) = b \times \{i\}$, set

$$W_{c(i)}^\parallel : \mathbb{Z}_N \rightarrow \mathbb{C}, \quad W_{c(i)}^\parallel(x) := \exp \left(\frac{\beta K_c}{M} \delta_N(x) \right) = 1 + \left(e^{\beta K_c / M} - 1 \right) \delta_N(x), \quad (2.21a)$$

$$V_{b(i)}^\parallel : \mathbb{Z}_N \rightarrow \mathbb{C}, \quad V_{b(i)}^\parallel(x) := \exp \left(\frac{\beta}{M} \sum_{n \in \mathbb{Z}_N} h_b^{(n)} \omega^{nx} \right). \quad (2.21b)$$

¹The sign in the 't Hooft operator was chosen so that it will result in a clean cocycle shift in the classical spacetime model based on our conventions for lattice differentials (see (2.27)).

For vertical P -cells $\underline{a}(i) = a \times [i, i+1]$ and vertical $(P+1)$ -cells $\underline{b}(i) = b \times [i, i+1]$, set

$$V_{\underline{a}(i)}^\perp : \mathbb{Z}_N \rightarrow \mathbb{C}, \quad V_{\underline{a}(i)}^\perp(x) := \delta_N(x) + \frac{e^{\beta J_a/M} - 1}{N}, \quad (2.22a)$$

$$W_{\underline{b}(i)}^\perp : \mathbb{Z}_N \rightarrow \mathbb{C}, \quad W_{\underline{b}(i)}^\perp(x) := \sum_{j \in \mathbb{Z}_N} \exp\left(\frac{\beta}{M} \sum_{n \in \mathbb{Z}_N} g_b^{(n)} \omega^{nj}\right) \frac{\omega^{(-1)^P jx}}{N}. \quad (2.22b)$$

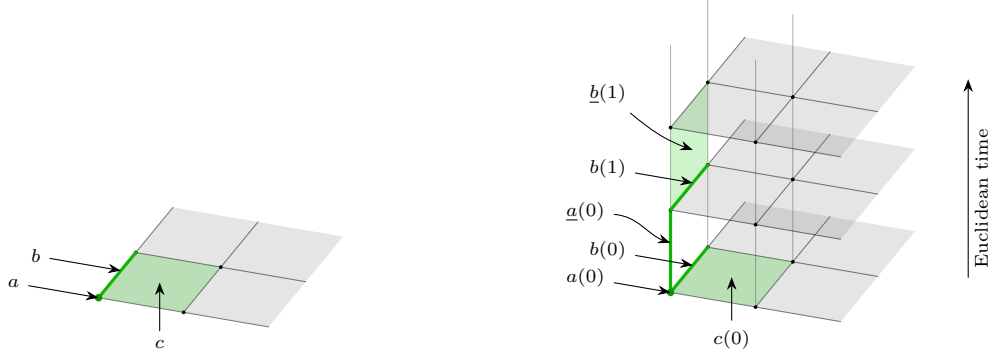
When no confusion is possible, we write simply W_c for either $W_c^\parallel$ or $W_c^\perp$, depending on whether $c \in C_{P+1}(\underline{\Lambda})$ is horizontal or vertical, and similarly V_u for $u \in C_P(\underline{\Lambda})$. Fig. 1 shows how cells of the spatial lattice transform into horizontal and vertical cells of the spacetime lattice.

Proposition 2.2 (Neutral thermal partition function). *The finite M partition function Z_M defined in (2.20) is given by*

$$Z_M = \sum_{\phi \in \Omega^P(\underline{\Lambda})} \prod_{c \in C_{P+1}(\underline{\Lambda})} W_c((D\phi)_c) \prod_{u \in C_P(\underline{\Lambda})} V_u(\phi_u). \quad (2.23)$$

Proof. This is (C.26), derived via the kernel computations in Appendix C.2. $\square$

Forms and differential on the spacetime lattice are defined in Appendix B.1.



(a) The spatial lattice Λ in the case $d = 2$, with a distinguished vertex $a \in C_0(\Lambda)$, edge $b \in C_1(\Lambda)$, and plaquette $c \in C_2(\Lambda)$.

(b) The spacetime lattice $\underline{\Lambda}$ obtained by stacking copies of Λ along the Euclidean time direction. The spatial cells a , b , and c give rise to horizontal cells $a(i)$, $b(i)$, and $c(i)$ in the i -th time slice, and to vertical cells $\underline{a}(i)$, $\underline{b}(i)$, and $\underline{c}(i)$ connecting adjacent slices.

Figure 1: Passage from the spatial square lattice Λ to the spacetime lattice $\underline{\Lambda}$ in the example $d = 2$, $P = 1$. All bold/green-shaded cells are emphasized only for visual clarity.

2.4 Spatial background

We first decorate the Trotterized trace by a spatial Wilson and 't Hooft insertion. Given

$$\nu \in Z_P(\Lambda), \quad \mu^\vee \in Z_{d-P}(\Lambda^\vee), \quad (2.24)$$

define

$$Z_M^{\text{sp}}(\mu^\vee, \nu) := \text{tr}_{\mathcal{H}} \left[e^{-\beta \hat{H}_z/M} \hat{\mathcal{W}}_\nu e^{-\beta \hat{H}_x/M} \hat{\mathcal{T}}_{\mu^\vee} \left(e^{-\beta \hat{H}_z/M} e^{-\beta \hat{H}_x/M} \right)^{M-1} \right]. \quad (2.25)$$

The Wilson operator is placed on the initial time slice, while the 't Hooft operator acts on the first slab.

Let $\nu(0) \in Z_P(\underline{\Lambda})$ denote the horizontal lift of ν to the initial time slice. Let $\mu^\vee(0) \in Z_{d-P}(\underline{\Lambda}^\vee)$ denote the dual cycle supported on the first dual slab and characterized by

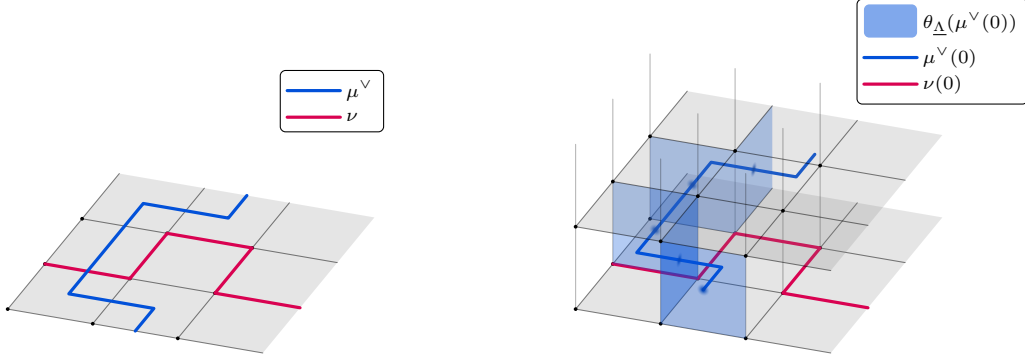
$$(\vartheta_{\underline{\Lambda}}(\mu^\vee(0)))_{\underline{b}(0)} = (\vartheta_{\Lambda}(\mu^\vee))_b, \quad (\vartheta_{\underline{\Lambda}}(\mu^\vee(0)))_{\underline{c}(i)} = 0, \quad (\vartheta_{\underline{\Lambda}}(\mu^\vee(0)))_{\underline{b}(i)} = 0 \quad (i \neq 0). \quad (2.26)$$

Fig. 2 illustrates how ν and $\mu^\vee$ are represented on the spacetime lattice.

Proposition 2.3 (Spatially decorated thermal partition function). *For every M , $\nu \in Z_P(\Lambda)$, and $\mu^\vee \in Z_{d-P}(\Lambda^\vee)$,*

$$Z_M^{\text{sp}}(\mu^\vee, \nu) = \sum_{\phi \in \Omega^P(\underline{\Lambda})} \chi \left(\int_{\nu(0)} \phi \right) \prod_{c \in C_{P+1}(\underline{\Lambda})} W_c((D\phi + \vartheta_{\underline{\Lambda}}(\mu^\vee(0)))_c) \prod_{u \in C_P(\underline{\Lambda})} V_u(\phi_u). \quad (2.27)$$

Proof. This is (C.34). The Wilson insertion gives the character $\chi(\int_{\nu(0)} \phi)$, while the 't Hooft insertion produces the vertical cocycle shift $\vartheta_{\underline{\Lambda}}(\mu^\vee(0))$. $\square$



(a) The spatial lattice Λ for the case $d = 2$, $P = 1$, with the supports $\nu \subset \Lambda$ and $\mu^\vee \subset \Lambda^\vee$ of the Wilson and 't Hooft operators inserted in the decorated spatial trace. The support ν is drawn on the primal lattice Λ , while the support $\mu^\vee$ consists of 1-cells of the dual lattice $\Lambda^\vee$.

(b) The corresponding spacetime supports in $\underline{\Lambda}$. The support $\nu(0) \subset \underline{\Lambda}$ of the Wilson insertion remains on the 0-th time slice, whereas the support $\mu^\vee(0) \subset \underline{\Lambda}^\vee$ of the 't Hooft insertion lies between the 0-th and 1-st slices. The blue-shaded plaquettes form the support of $\theta_{\underline{\Lambda}}(\mu^\vee(0))$, a 2-chain on the primal spacetime lattice defined by (B.29); equivalently, it consists of the primal plaquettes dual to the edges of $\mu^\vee(0)$. The 2-chain $\theta_{\underline{\Lambda}}(\mu^\vee(0))$ is the support of the 2-form $\vartheta_{\underline{\Lambda}}(\mu^\vee(0))$.

Figure 2: From spatial insertions to their spacetime representatives in the case $d = 2$, $P = 1$. The dual lattices $\Lambda^\vee$ and $\underline{\Lambda}^\vee$ are not shown.

2.5 Euclidean twists and generic background

We now introduce twists inserted uniformly around the Euclidean time circle. These generate spacetime backgrounds wrapping the thermal direction.

2.5.1 Electric twist.

Let

$$\alpha = \sum_{a \in C_{P-1}(\Lambda)} \alpha_a a \in Z_{P-1}(\Lambda). \quad (2.28)$$

For each $a \in C_{P-1}(\Lambda)$, define the spectral projectors of $\hat{A}_a$ by

$$\hat{\Pi}_{a,j}^{(A)} := \frac{1}{N} \sum_{m=0}^{N-1} \omega^{-jm} \hat{A}_a^m, \quad j \in \mathbb{Z}_N. \quad (2.29)$$

They satisfy

$$\hat{A}_a \hat{\Pi}_{a,j}^{(A)} = \omega^j \hat{\Pi}_{a,j}^{(A)}, \quad \sum_{j \in \mathbb{Z}_N} \hat{\Pi}_{a,j}^{(A)} = \hat{1}, \quad \hat{\mathcal{A}}_a = \hat{\Pi}_{a,0}^{(A)}. \quad (2.30)$$

Set

$$\lambda_{a,j} := 1 + \left(e^{\beta J_a / M} - 1 \right) \delta_N(j). \quad (2.31)$$

The electric twist operator is

$$\hat{U}_e(\alpha) := \prod_{a \in C_{P-1}(\Lambda)} \hat{U}_{e,a}(\alpha_a), \quad \hat{U}_{e,a}(\alpha_a) := \sum_{j \in \mathbb{Z}_N} \frac{\lambda_{a,j+\alpha_a}}{\lambda_{a,j}} \hat{\Pi}_{a,j}^{(A)}. \quad (2.32)$$

Its effect on the Trotter kernel is computed in Appendix C.4.1. It multiplies the i -th slab by

$$\prod_{a \in C_{P-1}(\Lambda)} \omega^{\alpha_a n(i)_a}. \quad (2.33)$$

Equivalently, after packaging into $\phi \in \Omega^P(\underline{\Lambda})$, the product over all slabs is the spacetime Wilson factor

$$\chi \left(\int_{\text{Sus}(\alpha)} \phi \right), \quad \text{Sus}(\alpha) := \sum_{i=0}^{M-1} \underline{\alpha}(i) \in Z_P(\underline{\Lambda}), \quad \underline{\alpha}(i) = \alpha \times [i, i+1]. \quad (2.34)$$

The total electric spacetime cycle is therefore

$$q_e(\nu, \alpha) := \nu(0) + \text{Sus}(\alpha) \in Z_P(\underline{\Lambda}). \quad (2.35)$$

2.5.2 Magnetic twist.

Let

$$\beta^\vee \in Z_{d-P-1}(\Lambda^\vee), \quad \beta_c := (\vartheta_\Lambda(\beta^\vee))_c, \quad c \in C_{P+1}(\Lambda). \quad (2.36)$$

For each $c \in C_{P+1}(\Lambda)$, define the spectral projectors of $\hat{B}_c$ by

$$\hat{\Pi}_{c,j}^{(B)} := \frac{1}{N} \sum_{m=0}^{N-1} \omega^{-jm} \hat{B}_c^m, \quad j \in \mathbb{Z}_N. \quad (2.37)$$

They satisfy

$$\hat{B}_c \hat{\Pi}_{c,j}^{(B)} = \omega^j \hat{\Pi}_{c,j}^{(B)}, \quad \sum_{j \in \mathbb{Z}_N} \hat{\Pi}_{c,j}^{(B)} = \hat{1}, \quad \hat{B}_c = \hat{\Pi}_{c,0}^{(B)}. \quad (2.38)$$

Set

$$\rho_{c,j} := 1 + \left(e^{\beta K_c/M} - 1 \right) \delta_N(j). \quad (2.39)$$

The magnetic twist operator is

$$\hat{U}_m(\beta^\vee) := \prod_{c \in C_{P+1}(\Lambda)} \hat{U}_{m,c}(\beta_c), \quad \hat{U}_{m,c}(\beta_c) := \sum_{j \in \mathbb{Z}_N} \frac{\rho_{c,j+\beta_c}}{\rho_{c,j}} \hat{\Pi}_{c,j}^{(B)}. \quad (2.40)$$

As shown in Appendix C.4.2, this shifts the horizontal curvature argument by $\vartheta_\Lambda(\beta^\vee)$. In spacetime terms, it produces the cocycle $\vartheta_{\underline{\Lambda}}(\text{Sus}(\beta^\vee))$, where

$$\text{Sus}(\beta^\vee) \in Z_{d-P}(\underline{\Lambda}^\vee) \quad (2.41)$$

is characterized by

$$(\vartheta_{\underline{\Lambda}}(\text{Sus}(\beta^\vee)))_{c(i)} = (\vartheta_\Lambda(\beta^\vee))_c, \quad (\vartheta_{\underline{\Lambda}}(\text{Sus}(\beta^\vee)))_{\underline{b}(i)} = 0. \quad (2.42)$$

Thus the total magnetic spacetime cycle is

$$q_m(\mu^\vee, \beta^\vee) := \mu^\vee(0) + \text{Sus}(\beta^\vee) \in Z_{d-P}(\underline{\Lambda}^\vee). \quad (2.43)$$

2.5.3 General background partition function

Define the fully decorated Trotterized trace by

$$\begin{aligned} Z_M(\mu^\vee, \nu; \beta^\vee, \alpha) := & \text{tr}_{\mathcal{H}} \left[\hat{\mathcal{W}}_\nu \hat{U}_m(\beta^\vee) e^{-\beta \hat{H}_z/M} \hat{\mathcal{T}}_{\mu^\vee} \hat{U}_e(\alpha) e^{-\beta \hat{H}_x/M} \right. \\ & \left. \times \left(\hat{U}_m(\beta^\vee) e^{-\beta \hat{H}_z/M} \hat{U}_e(\alpha) e^{-\beta \hat{H}_x/M} \right)^{M-1} \right]. \end{aligned} \quad (2.44)$$

Proposition 2.4 (General background partition function). *With $q_e(\nu, \alpha)$ and $q_m(\mu^\vee, \beta^\vee)$ defined by (2.35) and (2.43), the decorated trace (2.44) is exactly*

$$Z_M(\mu^\vee, \nu; \beta^\vee, \alpha) = \sum_{\phi \in \Omega^P(\underline{\Lambda})} \chi \left(\int_{q_e(\nu, \alpha)} \phi \right) \prod_{c \in C_{P+1}(\underline{\Lambda})} W_c((D\phi + \vartheta_{\underline{\Lambda}}(q_m(\mu^\vee, \beta^\vee)))_c) \prod_{u \in C_P(\underline{\Lambda})} V_u(\phi_u). \quad (2.45)$$

Proof. The spatial Wilson insertion and the electric Euclidean twist combine into the single character $\chi \left(\int_{q_e(\nu, \alpha)} \phi \right)$. The spatial 't Hooft insertion and the magnetic Euclidean twist combine into the single cocycle shift $\vartheta_{\underline{\Lambda}}(q_m(\mu^\vee, \beta^\vee))$. The detailed verification of these two statements is the content of Appendix C.4, C.5. Substituting the resulting electric factor and magnetic cocycle shift into the neutral spacetime model (2.23) gives (2.45). $\square$

Proposition 2.4 is exact at fixed Trotter number M . The physical thermal partition function of the quantum Hamiltonian is recovered only after taking the Trotter limit $M \rightarrow \infty$.

Remark 2.5 (The trace in a different basis). Instead of computing the Trotterized trace in the Z -basis as we do, one can also compute it in the X -basis, obtaining a dual description of the classical theory with interaction defined by the transposed differential $D^\top$. For the special case of the $\mathbb{Z}_2$ toric code on a regular 2d square lattice with spins on edges, and without any decorations, this route was taken in [39].

3 Polymer gas in topological backgrounds

In this section we rewrite the exact spacetime partition function (2.45) as a gas of extended magnetic and electric defects. The resulting representation is exact in finite volume and is naturally viewed as an expansion around the flat-field vacuum. The section has three main outputs.

First, we recast the decorated spacetime partition function in terms of arbitrary magnetic and electric spacetime cycles. Second, after expanding the local weights, we show that the sum over the spacetime field forces the local defects to assemble into closed magnetic and electric defects whose total homology classes are fixed by the chosen background. Third, we resolve the resulting homology constraints by finite Fourier transform, obtaining a sectorwise polymer gas indexed by topological charges.

In §2, the electric and magnetic backgrounds were produced from a combination of spatial cycles and cycles wrapping the Euclidean-time direction. Since topologically $\underline{\Lambda} = \Lambda \times S^1$ and $\underline{\Lambda}^\vee = \Lambda^\vee \times S^1$, the Künneth decomposition gives

$$H_P(\underline{\Lambda}) \cong H_P(\Lambda) \oplus H_{P-1}(\Lambda), \quad H_{d-P}(\underline{\Lambda}^\vee) \cong H_{d-P}(\Lambda^\vee) \oplus H_{d-P-1}(\Lambda^\vee). \quad (3.1)$$

The cycles $q_e(\nu, \alpha)$ (2.35) and $q_m(\mu^\vee, \beta^\vee)$ (2.43) are specific representatives of the homology classes of the primal and the dual lattice compatible with the above decomposition. While working with the classical theory, it will be more convenient to work with arbitrary spacetime representatives

$$q_m \in Z_{d-P}(\underline{\Lambda}^\vee), \quad q_e \in Z_P(\underline{\Lambda}). \quad (3.2)$$

We therefore define the fixed-charge partition function by

$$Z_M(q_m, q_e) := \sum_{\phi \in \Omega^P(\underline{\Lambda})} \chi \left(\int_{q_e} \phi \right) \prod_{c \in C_{P+1}(\underline{\Lambda})} W_c((D\phi + \vartheta_{\underline{\Lambda}}(q_m))_c) \prod_{u \in C_P(\underline{\Lambda})} V_u(\phi_u), \quad (3.3)$$

where q_e is a primal P -cycle, q_m is a dual $(d-P)$ -cycle, and $\vartheta_{\underline{\Lambda}}(q_m) \in Z^{P+1}(\underline{\Lambda})$ (see (B.33)) is the corresponding primal cocycle.

3.1 Code limit and the flat vacuum

For a perturbative expansion of Trotter weights (2.21), (2.22), it is convenient at fixed M to introduce

$$\epsilon_g := \max_{b \in C_P(\Lambda)} \sum_{n \in \mathbb{Z}_N} \left| \frac{\beta g_b^{(n)}}{M} \right|, \quad \epsilon_h := \max_{b \in C_P(\Lambda)} \sum_{n \in \mathbb{Z}_N} \left| \frac{\beta h_b^{(n)}}{M} \right|, \quad (3.4)$$

and

$$\kappa_J := \min_{a \in C_{P-1}(\Lambda)} \frac{\beta J_a}{M}, \quad \kappa_K := \min_{c \in C_{P+1}(\Lambda)} \frac{\beta K_c}{M}. \quad (3.5)$$

The *code limit* is the regime

$$\epsilon_g \rightarrow 0, \quad \epsilon_h \rightarrow 0, \quad \kappa_J \rightarrow \infty, \quad \kappa_K \rightarrow \infty. \quad (3.6)$$

At the level of the explicit local weights (2.21), (2.22), this means that, after dividing out the trivial local zero-mode factors,

$$\frac{W_c(x)}{W_c(0)} \rightarrow \delta_N(x), \quad \frac{V_u(x)}{V_u(0)} \rightarrow 1, \quad (3.7)$$

uniformly in $x \in \mathbb{Z}_N$. Accordingly, define

$$\tilde{Z}_M(q_m, q_e) := \frac{Z_M(q_m, q_e)}{\prod_{c \in C_{P+1}(\underline{\Lambda})} W_c(0) \prod_{u \in C_P(\underline{\Lambda})} V_u(0)}. \quad (3.8)$$

Then in the code limit

$$\tilde{Z}_M(q_m, q_e) \rightarrow \sum_{\phi \in \Omega^P(\underline{\Lambda})} \chi \left(\int_{q_e} \phi \right) \prod_{c \in C_{P+1}(\underline{\Lambda})} \delta_N((D\phi + \vartheta_{\underline{\Lambda}}(q_m))_c). \quad (3.9)$$

In the neutral magnetic sector $q_m = 0$, the limiting theory is therefore supported on the flat sector

$$D\phi = 0. \quad (3.10)$$

This is the *flat vacuum* around which the polymer expansion will be organized. The defect variables introduced below measure departures from this flat-field configuration, and in the neutral sector the dominant defect configuration is the empty one.

3.2 Local defect expansion

We now expand (3.3) around the flat vacuum by separating the local zero modes from the nontrivial local excitations.

For each $(P+1)$ -cell $c \in C_{P+1}(\underline{\Lambda})$, define

$$\varpi_c(x) := \frac{W_c(x)}{W_c(0)}, \quad x \in \mathbb{Z}_N. \quad (3.11)$$

Then $\varpi_c(0) = 1$, and

$$W_c(x) = W_c(0) \sum_{m \in \mathbb{Z}_N} \varpi_c(m) \delta_N(x - m). \quad (3.12)$$

A nonzero m will be interpreted as a local magnetic defect on the cell c .

For each P -cell $u \in C_P(\underline{\Lambda})$, let

$$\hat{V}_u(e) := \frac{1}{\sqrt{N}} \sum_{x \in \mathbb{Z}_N} V_u(x) \chi_e(-x), \quad e \in \hat{\mathbb{Z}}_N, \quad (3.13)$$

be the discrete Fourier transform in the convention of (A.6b). Define the normalized electric Fourier amplitudes by

$$v_u(e) := \frac{\hat{V}_u(-e)}{\hat{V}_u(0)}, \quad e \in \mathbb{Z}_N. \quad (3.14)$$

Then $v_u(0) = 1$, and Fourier inversion gives

$$V_u(x) = \frac{\hat{V}_u(0)}{\sqrt{N}} \sum_{e \in \mathbb{Z}_N} v_u(e) \chi_e(-x). \quad (3.15)$$

A nonzero e will be interpreted as a local electric defect label on the cell u .

In the code limit (3.6), the nontrivial amplitudes are small:

$$\varpi_c(m) \rightarrow 0 \quad (m \neq 0), \quad v_u(e) \rightarrow 0 \quad (e \neq 0). \quad (3.16)$$

This is the precise sense in which the polymer gas below is a perturbation series around the flat vacuum.

Introduce magnetic and electric defect chains

$$\mathbf{m}^\vee = \sum_{c^\vee \in C_{d-P}(\underline{\Lambda}^\vee)} \mathbf{m}_{c^\vee}^\vee c^\vee \in C_{d-P}(\underline{\Lambda}^\vee), \quad \mathbf{e} = \sum_{u \in C_P(\underline{\Lambda})} \mathbf{e}_u u \in C_P(\underline{\Lambda}). \quad (3.17)$$

Let

$$\mathbf{m} := \vartheta_{\underline{\Lambda}}(\mathbf{m}^\vee) \in \Omega^{P+1}(\underline{\Lambda}), \quad \mathbf{m}_c = \mathbf{m}_{\vartheta_{\underline{\Lambda}}^{-1}(c)}^\vee \quad (c \in C_{P+1}(\underline{\Lambda})), \quad (3.18)$$

and define the corresponding defect weights by

$$\text{wt}_m(\mathbf{m}^\vee) := \prod_{c \in C_{P+1}(\underline{\Lambda})} \varpi_c(\mathbf{m}_c), \quad \text{wt}_e(\mathbf{e}) := \prod_{u \in C_P(\underline{\Lambda})} v_u(\mathbf{e}_u). \quad (3.19)$$

Finally, set

$$\mathcal{N}_M := \prod_{c \in C_{P+1}(\underline{\Lambda})} W_c(0) \prod_{u \in C_P(\underline{\Lambda})} \frac{\widehat{V}_u(0)}{\sqrt{N}}. \quad (3.20)$$

Proposition 3.1 (Exact local-defect expansion). *For every magnetic background $q_m \in Z_{d-P}(\underline{\Lambda}^\vee)$ and electric background $q_e \in Z_P(\underline{\Lambda})$, the partition function (3.3) is given by*

$$Z_M(q_m, q_e) = \mathcal{N}_M \sum_{\mathbf{m}^\vee \in C_{d-P}(\underline{\Lambda}^\vee)} \sum_{\mathbf{e} \in C_P(\underline{\Lambda})} \text{wt}_m(\mathbf{m}^\vee) \text{wt}_e(\mathbf{e}) \Xi(\mathbf{m}^\vee, \mathbf{e}; q_m, q_e), \quad (3.21)$$

where

$$\Xi(\mathbf{m}^\vee, \mathbf{e}; q_m, q_e) := \sum_{\phi \in \Omega^P(\underline{\Lambda})} \chi \left(\int_{q_e - \mathbf{e}} \phi \right) \prod_{c \in C_{P+1}(\underline{\Lambda})} \delta_N((D\phi + \vartheta_{\underline{\Lambda}}(q_m - \mathbf{m}^\vee))_c). \quad (3.22)$$

Proof. Insert (3.12) and (3.15) into (3.3). The factors $W_c(0)$ and $\widehat{V}_u(0)/\sqrt{N}$ assemble into $\mathcal{N}_M$, while the remaining cellwise products produce the defect weights (3.19). The magnetic defect variables are naturally assembled into the dual chain $\mathbf{m}^\vee$, whose primal image is the cochain $\mathbf{m}$. The electric Fourier labels assemble into the primal chain $\mathbf{e}$. Reordering the finite sums then gives (3.21) with compatibility factor (3.22). $\square$

Equation (3.22) exhibits the basic structure of the expansion. The chains $\mathbf{m}^\vee$ and $\mathbf{e}$ are local magnetic and electric defect fields, while $\Xi(\mathbf{m}^\vee, \mathbf{e}; q_m, q_e)$ is the compatibility factor obtained by summing over all spacetime P -cochains consistent with those local defects in the prescribed background.

3.3 Closed defect expansion and homology constraints

We now determine when the compatibility factor $\Xi(\mathbf{m}^\vee, \mathbf{e}; q_m, q_e)$ is nonzero.

Suppose first that the delta constraints in (3.22) are solvable. Then there exists $\phi \in \Omega^P(\underline{\Lambda})$ such that

$$D\phi = \vartheta_{\underline{\Lambda}}(\mathbf{m}^\vee - q_m). \quad (3.23)$$

Since the left-hand side is a coboundary and $\vartheta_{\underline{\Lambda}}$ is an isomorphism (Corr. B.2), it follows that

$$\mathbf{m}^\vee \in Z_{d-P}(\underline{\Lambda}^\vee), \quad [\mathbf{m}^\vee] = [q_m] \in H_{d-P}(\underline{\Lambda}^\vee), \quad (3.24)$$

because (3.23) is precisely the statement that $\mathbf{m}^\vee - q_m \in B_{d-P}(\underline{\Lambda}^\vee)$.

Next fix one solution λ of (3.23). Every other solution is of the form $\lambda + \psi$ with $\psi \in Z^P(\underline{\Lambda})$. Therefore

$$\Xi(\mathbf{m}^\vee, \mathbf{e}; q_m, q_e) = \chi \left(\int_{q_e - \mathbf{e}} \lambda \right) \sum_{\psi \in Z^P(\underline{\Lambda})} \chi \left(\int_{q_e - \mathbf{e}} \psi \right). \quad (3.25)$$

By character orthogonality on the finite group of cocycles, this vanishes unless $q_e - \mathfrak{e}$ pairs trivially with every cocycle, that is, unless

$$q_e - \mathfrak{e} \in B_P(\underline{\Lambda}). \quad (3.26)$$

Equivalently,

$$\mathfrak{e} \in Z_P(\underline{\Lambda}), \quad [\mathfrak{e}] = [q_e] \in H_P(\underline{\Lambda}). \quad (3.27)$$

Thus only closed magnetic and electric defects survive, and their total homology classes are forced to agree with those of the prescribed background. Since $q_e - \mathfrak{e}$ is a boundary, the character $\chi \left(\int_{q_e - \mathfrak{e}} \psi \right)$ with closed ψ evaluates to 1 and the sum over ψ contributes the factor $|Z^P(\underline{\Lambda})|$. It is therefore convenient to define the normalized partition function

$$\mathcal{Z}_M(q_m, q_e) := \frac{Z_M(q_m, q_e)}{\mathcal{N}_M |Z^P(\underline{\Lambda})|}. \quad (3.28)$$

The remaining phase is determined by the two boundary differences

$$\delta_m := \mathbf{m}^\vee - q_m \in B_{d-P}(\underline{\Lambda}^\vee), \quad \delta_e := q_e - \mathfrak{e} \in B_P(\underline{\Lambda}). \quad (3.29)$$

Let $\Sigma \in \mathcal{C}_{P+1}(\underline{\Lambda})$ such that $\partial\Sigma = \delta_e$, then the Stokes formula gives²

$$\int_{\delta_e} \lambda = \int_{\partial\Sigma} \lambda = \int_{\Sigma} D\lambda = I_{\underline{\Lambda}}(\delta_m, \Sigma). \quad (3.30)$$

This is the first term in the definition of the linking number between δ_m and δ_e (B.55), and since both of these are boundaries, the remaining two terms in (B.55) vanish. Hence

$$\chi \left(\int_{q_e - \mathfrak{e}} \lambda \right) = \omega^{\text{Lk}_{\underline{\Lambda}}(\mathbf{m}^\vee - q_m, q_e - \mathfrak{e})}. \quad (3.31)$$

Proposition 3.2 (Closed-defect gas at fixed background). *The normalized partition function (3.28) is given by*

$$\mathcal{Z}_M(q_m, q_e) = \sum_{\substack{\mathbf{m}^\vee \in Z_{d-P}(\underline{\Lambda}^\vee) \\ [\mathbf{m}^\vee] = [q_m]}} \sum_{\substack{\mathfrak{e} \in Z_P(\underline{\Lambda}) \\ [\mathfrak{e}] = [q_e]}} \text{wt}_m(\mathbf{m}^\vee) \text{wt}_e(\mathfrak{e}) \omega^{-\text{Lk}_{\underline{\Lambda}}(\mathbf{m}^\vee - q_m, \mathfrak{e} - q_e)}. \quad (3.32)$$

Proof. The preceding discussion shows that $\Xi(\mathbf{m}^\vee, \mathfrak{e}; q_m, q_e)$ vanishes unless $[\mathbf{m}^\vee] = [q_m]$ and $[\mathfrak{e}] = [q_e]$. In the nonvanishing case, the cocycle sum in (3.25) contributes $|Z^P(\underline{\Lambda})|$, and the remaining character is (3.31). Substituting this character and $|Z^P(\underline{\Lambda})|$ into (3.21) and normalizing as in (3.28) gives (3.32). $\square$

The above proposition holds for every integer $N \geq 2$ since the phase in (3.32) involves only the boundary differences so that it is the ordinary canonical linking number of two boundaries and does not require N to be prime (Prop. B.3(ii)). In the rest of §3 we will use a chosen bilinear extension of that boundary-linking to arbitrary cycles, first in the bilinear expansion (3.43) and then in the Fourier resolution of the homology constraints. For those steps we assume N is prime, so that $\mathbb{Z}_N = \mathbb{F}_N$ and the homology groups are finite-dimensional $\mathbb{F}_N$ -vector spaces.

3.4 Connected polymer expansion

A nonzero closed defect need not be connected. We therefore decompose each surviving magnetic and electric defect into its support-connected components:

$$\mathbf{m}^\vee = M_1 + \cdots + M_r, \quad \mathfrak{e} = E_1 + \cdots + E_s, \quad (3.33)$$

where the supports of the M_i are pairwise disjoint, the supports of the E_j are pairwise disjoint, and

$$M_i \in Z_{d-P}(\underline{\Lambda}^\vee), \quad E_j \in Z_P(\underline{\Lambda}). \quad (3.34)$$

²In fact, we define the lattice differential to satisfy this condition, see (B.10) for the spatial lattice, applies more generally.

Define

$$\begin{aligned}\mathbb{M} &:= \{\text{connected } (d-P)\text{-cycles in } \underline{\Lambda}^\vee\}, \\ \mathbb{E} &:= \{\text{connected } P\text{-cycles in } \underline{\Lambda}\}.\end{aligned}\tag{3.35}$$

We refer to elements of $\mathbb{M}$ as magnetic polymers and to elements of $\mathbb{E}$ as electric polymers.

Because the defect weights factor over cells, they factor over support-disjoint unions. Thus there exist one-polymer activities

$$\rho_m : \mathbb{M} \rightarrow \mathbb{C}, \quad \rho_e : \mathbb{E} \rightarrow \mathbb{C}\tag{3.36}$$

such that

$$\text{wt}_m(\mathbf{m}^\vee) = \prod_{i=1}^r \rho_m(M_i), \quad \text{wt}_e(\mathbf{e}) = \prod_{j=1}^s \rho_e(E_j).\tag{3.37}$$

Explicitly,

$$\rho_m(M) = \prod_{c \in C_{P+1}(\underline{\Lambda})} \varpi_c((\vartheta_{\underline{\Lambda}}(M))_c), \quad \rho_e(E) = \prod_{u \in C_P(\underline{\Lambda})} v_u(E_u).\tag{3.38}$$

Introduce the homology groups

$$H_m := H_{d-P}(\underline{\Lambda}^\vee), \quad H_e := H_P(\underline{\Lambda}).\tag{3.39}$$

The groups H_m and H_e label the topological classes of the magnetic and electric background charges respectively, i.e., they are the groups in which the homology classes $[q_m]$ and $[q_e]$ take values.

Let $\delta_H(h)$ be the Kronecker delta on any abelian group H . Substituting (3.33) and (3.37) into (3.32) gives the fixed-background polymer gas:

$$\begin{aligned}\mathcal{Z}_M(q_m, q_e) &= \sum_{r,s \geq 0} \frac{1}{r! s!} \sum_{\substack{M_1, \dots, M_r \in \mathbb{M} \\ \text{pairwise disjoint}}} \sum_{\substack{E_1, \dots, E_s \in \mathbb{E} \\ \text{pairwise disjoint}}} \left(\prod_{i=1}^r \rho_m(M_i) \right) \left(\prod_{j=1}^s \rho_e(E_j) \right) \\ &\quad \times \delta_{H_m} \left(\sum_{i=1}^r [M_i] - [q_m] \right) \delta_{H_e} \left(\sum_{j=1}^s [E_j] - [q_e] \right) \omega^{-\text{Lk}_{\underline{\Lambda}} \left(\sum_{i=1}^r M_i - q_m, \sum_{j=1}^s E_j - q_e \right)}.\end{aligned}\tag{3.40}$$

This is an exact perturbation series around the code limit. The local defect amplitudes $\varpi_c(m)$ and $v_u(e)$ measure departures from the flat vacuum; the ϕ -sum forces these local excitations to assemble into closed extended objects; and their connected components are the polymers. The background data enters in two distinct ways. First, it imposes the homology constraints

$$\sum_i [M_i] = [q_m], \quad \sum_j [E_j] = [q_e].\tag{3.41}$$

Second, after these constraints hold, the surviving character is the linking phase of the boundary differences

$$\sum_i M_i - q_m \in B_{d-P}(\underline{\Lambda}^\vee), \quad \sum_j E_j - q_e \in B_P(\underline{\Lambda}).\tag{3.42}$$

Equivalently, by bilinearity,

$$\begin{aligned}& -\text{Lk}_{\underline{\Lambda}} \left(\sum_i M_i - q_m, \sum_j E_j - q_e \right) \\ &= -\sum_{i,j} \text{Lk}_{\underline{\Lambda}}(M_i, E_j) + \sum_i \text{Lk}_{\underline{\Lambda}}(M_i, q_e) + \sum_j \text{Lk}_{\underline{\Lambda}}(q_m, E_j) - \text{Lk}_{\underline{\Lambda}}(q_m, q_e).\end{aligned}\tag{3.43}$$

Thus the exact phase contains the polymer-polymer interaction, the interaction of polymers with the chosen background representatives, and a background-background constant.³ In the neutral sector, the empty polymer configuration is the vacuum term. In a nontrivial fixed-background sector, the leading contribution comes from the smallest polymer configurations carrying the required homology.

³Note that the individual terms on the right-hand side of (3.43) involve the chosen generalized linking pairing on arbitrary cycles and therefore depend on the auxiliary conventions fixed in Appendix B.4. Only the total combination, which is the linking of the two boundary differences appearing on the left-hand side, is canonical.

3.5 Fourier resolution of the homology constraints

We can resolve the delta constraint in the partition function (3.40) and write it as a sum over two-species polymer gas partition functions. Each two-species partition function will be labeled by the *topological sector* data $(\ell_m, \ell_e) \in \widehat{H}_m \times \widehat{H}_e$ where $\widehat{H}_m$ and $\widehat{H}_e$ are the linear duals of the $\mathbb{F}_N$ -vector spaces H_m and H_e respectively. Given the background charges $(q_m, q_e) \in Z_{d-P}(\underline{\Lambda}^\vee) \times Z_P(\underline{\Lambda})$ and for any $(\ell_m, \ell_e) \in \widehat{H}_m \times \widehat{H}_e$ define the 1-polymer activity

$$\rho^{\ell_m, \ell_e; q_m, q_e}(A) := \begin{cases} \rho_m(A) \omega^{\ell_m([A]) + \text{Lk}_{\underline{\Lambda}}(M, q_e)}, & A \in \mathbb{M}, \\ \rho_e(A) \omega^{\ell_e([A]) + \text{Lk}_{\underline{\Lambda}}(q_m, E)}, & A \in \mathbb{E}. \end{cases} \quad (3.44)$$

This is a modification of the activity in (3.40) by the (ℓ_m, ℓ_e) -dependent phase.

Define an interaction between electric and magnetic polymer pairs as follows. Same-species polymers have hard-core exclusion: two magnetic polymers are compatible only when their supports are disjoint, and similarly for two electric polymers. Opposite-species polymers have no hard-core exclusion; instead, a magnetic polymer M and an electric polymer E interact through the linking phase $\omega^{-\text{Lk}_{\underline{\Lambda}}(M, E)}$. Equivalently, set

$$\zeta(A, B) := \begin{cases} -1, & A, B \text{ same species with overlapping supports,} \\ 0, & A, B \text{ same species with disjoint supports,} \\ \omega^{-\text{Lk}_{\underline{\Lambda}}(A, B)} - 1, & A \in \mathbb{M}, B \in \mathbb{E}, \\ \omega^{-\text{Lk}_{\underline{\Lambda}}(B, A)} - 1, & A \in \mathbb{E}, B \in \mathbb{M}. \end{cases} \quad (3.45)$$

Then

$$1 + \zeta(M, E) = \omega^{-\text{Lk}_{\underline{\Lambda}}(M, E)} \quad (3.46)$$

for opposite-species pairs, while $1 + \zeta(A, B)$ enforces hard-core exclusion for overlapping same-species pairs.

Proposition 3.3 (Two-species polymer gas). *The normalized fixed-background partition function (3.40) decomposes as a sum over magnetic and electric topological sectors*

$$\mathcal{Z}_M(q_m, q_e) = \frac{1}{|\widehat{H}_m| |\widehat{H}_e|} \sum_{\ell_m \in \widehat{H}_m} \sum_{\ell_e \in \widehat{H}_e} \omega^{-\ell_m([q_m]) - \ell_e([q_e])} \widehat{\mathcal{Z}}_M(\ell_m, \ell_e; q_m, q_e). \quad (3.47)$$

For each fixed $(\ell_m, \ell_e; q_m, q_e)$, the sector amplitude is the abstract two-species polymer partition function

$$\widehat{\mathcal{Z}}_M(\ell_m, \ell_e; q_m, q_e) = \omega^{-\text{Lk}_{\underline{\Lambda}}(q_m, q_e)} \sum_{n \geq 0} \frac{1}{n!} \sum_{A_1, \dots, A_n \in \mathbb{A}} \prod_{k=1}^n \rho^{\ell_m, \ell_e; q_m, q_e}(A_k) \prod_{1 \leq k < l \leq n} (1 + \zeta(A_k, A_l)). \quad (3.48)$$

Proof. The delta functions in (3.40) can be resolved by using

$$\delta_H(h) = \frac{1}{|\widehat{H}|} \sum_{\ell \in \widehat{H}} \omega^{\ell(h)}, \quad (3.49)$$

The resolution holds for any finite $\mathbb{F}_N$ -vector space H , and in particular, for H_m and H_e . Using (3.49) in (3.40) produces the factor $|\widehat{H}_m|^{-1} |\widehat{H}_e|^{-1} \omega^{-\ell_m([q_m]) - \ell_e([q_e])}$ and inserts

$$\prod_i \omega^{\ell_m([M_i])}, \quad \prod_j \omega^{\ell_e([E_j])}$$

into the magnetic and electric polymer weights. These are the (ℓ_m, ℓ_e) -dependent phases in (3.44).

Next expand the linking phase in (3.40) by bilinearity, as in (3.43). The polymer-background terms are precisely those absorbed into (3.44). The background-background term gives the overall phase $\omega^{-\text{Lk}_{\underline{\Lambda}}(q_m, q_e)}$. The polymer-polymer terms give the opposite-species pair factors encoded by (3.46).

It remains only to pass from the separated magnetic/electric sum to the abstract one-set notation. Same-species overlap is exactly the hard-core exclusion encoded by the first two cases of (3.45), while

opposite-species pairs contribute the linking phase (3.46). Finally, summing over ordered n -tuples in $\mathbb{A} = \mathbb{M} \sqcup \mathbb{E}$ with the factor $1/n!$ is equivalent to summing over r magnetic and s electric polymers with the factor $1/(r!s!)$, since

$$\frac{1}{n!} \binom{n}{r} = \frac{1}{r!s!}.$$

This proves (3.47) and (3.48). $\square$

Thus the Fourier transform over homology separates the fixed-background polymer gas into sector amplitudes labelled by (ℓ_m, ℓ_e) . In each topological sector, the dependence on the background representatives is split into a constant background-background phase and background-twisted one-polymer activities, while the genuine two-body interaction is the same electric-magnetic linking phase as in the neutral theory.

4 A rigorous low-activity region

The purpose of this section is to obtain rigorous control of the background-dependent fixed-charge partition functions $\mathcal{Z}_M(q_m, q_e)$ in the low-activity regime. The key idea is to compare the absolute value of the complex polymer expansion to positive same-species hard-core polymer gases and then apply abstract polymer criteria due to Kotecký-Preiss, Fernández-Procacci, Bissacot-Fernández-Procacci, and related cluster-expansion methods [20–23].

When the Fourier-resolved representation (3.48) is available, the corresponding sector amplitudes $\widehat{\mathcal{Z}}_M(\ell_m, \ell_e; q_m, q_e)$ have generally complex one-polymer activities (3.44), and opposite-species polymers interact through the pure phase (3.46). In our present formulation this sector decomposition is available for prime N , and in that case we first bound the sector amplitudes and then deduce bounds for the background-dependent partition functions $\mathcal{Z}_M(q_m, q_e)$. However, as explained in Remark 4.4, the same fixed-background partition functions can also be bounded directly, without passing through the sector amplitudes. The outcome is a background-uniform low-activity region in which the partition functions are absolutely controlled uniformly in finite volume, large connected polymers are exponentially suppressed, and homologically nontrivial polymers are exponentially suppressed on the scale of the corresponding spacetime systole, i.e. the minimal support size of a nontrivial homology cycle. The main limitation in the resulting region comes not from the abstract polymer criterion itself, but from the available combinatorial bound on the number of rooted connected closed polymers.

The prime- N assumption enters this section only through the sector decomposition (3.48). The positive majorant gases and the resulting low-activity bounds for the fixed-background partition functions $\mathcal{Z}_M(q_m, q_e)$ apply for every integer $N \geq 2$.

4.1 Sectorwise gas and absolute majorization

Fix background representatives

$$q_m \in Z_{d-P}(\underline{\Lambda}^\vee), \quad q_e \in Z_P(\underline{\Lambda}), \quad (4.1)$$

and topological sector charges

$$(\ell_m, \ell_e) \in \widehat{H}_m \times \widehat{H}_e. \quad (4.2)$$

We use the two-species polymer representation (3.48). Thus the polymer set is $\mathbb{A} = \mathbb{M} \sqcup \mathbb{E}$, the background-twisted one-polymer activity is $\rho^{\ell_m, \ell_e; q_m, q_e}$, and the pair interaction is ζ from (3.45).

The important observation is that all background and character dependences in the one-polymer activities are phases. Hence

$$|\rho^{\ell_m, \ell_e; q_m, q_e}(M)| = |\rho_m(M)|, \quad |\rho^{\ell_m, \ell_e; q_m, q_e}(E)| = |\rho_e(E)|. \quad (4.3)$$

Likewise, the overall background phase $\omega^{-\text{Lk}_{\underline{\Lambda}}(q_m, q_e)}$ has modulus one, and for opposite-species pairs

$$|1 + \zeta(M, E)| = \left| \omega^{-\text{Lk}_{\underline{\Lambda}}(M, E)} \right| = 1. \quad (4.4)$$

For same-species pairs, $1 + \zeta(A, B)$ is either 0 or 1, depending on whether the two polymers overlap or are support-disjoint. Thus taking absolute values removes all electric-magnetic phases and retains

only same-species hard-core exclusion. This allows us to use the Fernández-Procacci criterion [21] for a hard-core polymer gas. Instead of applying this criterion to the complex polymer gas, we use the fact that the absolute value of the two-species expansion is bounded by a product of two positive one-species hard-core gases.

For $s \in \{m, e\}$, write

$$\mathbb{A}_m := \mathbb{M}, \quad \mathbb{A}_e := \mathbb{E}. \quad (4.5)$$

The associated positive hard-core polymer gas of species s is

$$\mathcal{Z}_s^{\text{maj}} := \sum_{n \geq 0} \frac{1}{n!} \sum_{A_1, \dots, A_n \in \mathbb{A}_s} \left(\prod_{k=1}^n |\rho_s(A_k)| \right) \prod_{1 \leq k < l \leq n} \mathbf{1}\{A_k \sim A_l\}, \quad (4.6)$$

where same-species compatibility means support-disjointness.

For every fixed $(\ell_m, \ell_e; q_m, q_e)$, the absolute value of the sectorwise expansion satisfies

$$\left| \widehat{\mathcal{Z}}_M(\ell_m, \ell_e; q_m, q_e) \right| \leq \mathcal{Z}_m^{\text{maj}} \mathcal{Z}_e^{\text{maj}}. \quad (4.7)$$

Indeed, after absolute values are taken, magnetic and electric polymers no longer interact with one another, while same-species hard-core constraints remain. Consequently, absolute convergence of the two positive gases $\mathcal{Z}_m^{\text{maj}}$ and $\mathcal{Z}_e^{\text{maj}}$ implies absolute convergence, uniformly in $(\ell_m, \ell_e; q_m, q_e)$, of the sectorwise amplitudes $\widehat{\mathcal{Z}}_M(\ell_m, \ell_e; q_m, q_e)$.

Since the fixed-background amplitudes are recovered by the finite sum (3.47), the same majorant controls $\mathcal{Z}_M(q_m, q_e)$. Explicitly,

$$|\mathcal{Z}_M(q_m, q_e)| \leq \frac{1}{|\widehat{H}_m| |\widehat{H}_e|} \sum_{\ell_m, \ell_e} \left| \widehat{\mathcal{Z}}_M(\ell_m, \ell_e; q_m, q_e) \right| \leq \mathcal{Z}_m^{\text{maj}} \mathcal{Z}_e^{\text{maj}}. \quad (4.8)$$

Whenever $\mathcal{Z}_s^{\text{maj}}$ is finite and nonzero, define the pinned one-polymer amplitude of the positive gas by

$$\Pi_s^{\text{maj}}(A) := \frac{\partial}{\partial |\rho_s(A)|} \log \mathcal{Z}_s^{\text{maj}}, \quad A \in \mathbb{A}_s. \quad (4.9)$$

Equivalently,

$$|\rho_s(A)| \Pi_s^{\text{maj}}(A) = \frac{|\rho_s(A)|}{\mathcal{Z}_s^{\text{maj}}} \frac{\partial}{\partial |\rho_s(A)|} \mathcal{Z}_s^{\text{maj}}. \quad (4.10)$$

This is the normalized positive weight of configurations containing A . In the original complex sectorwise gas, the analogous quantity is not a probability. The majorant quantity (4.10) is the positive envelope that will control polymer tails.

4.2 Activity and counting bounds

We now formulate the two model-dependent inputs needed for convergence in large volume.

For magnetic polymers, define the carrier support on primal $(P+1)$ -cells by

$$\text{car}(M) := \theta_{\underline{\Lambda}}(\text{Supp}(M)) \subset C_{P+1}(\underline{\Lambda}), \quad |M| := |\text{car}(M)|, \quad M \in \mathbb{M}. \quad (4.11)$$

For electric polymers, set

$$\text{car}(E) := \text{Supp}(E) \subset C_P(\underline{\Lambda}), \quad |E| := |\text{Supp}(E)|, \quad E \in \mathbb{E}. \quad (4.12)$$

Define the carrier-cell sets

$$\mathfrak{C}_m := C_{P+1}(\underline{\Lambda}), \quad \mathfrak{C}_e := C_P(\underline{\Lambda}). \quad (4.13)$$

Thus, for $A \in \mathbb{A}_s$, the statement $u \in \text{car}(A)$ is meaningful for $u \in \mathfrak{C}_s$.

Define the uniform nontrivial local amplitudes by

$$t_m := \max_{\substack{c \in C_{P+1}(\underline{\Lambda}) \\ m \in \mathbb{Z}_N \setminus \{0\}}} |\varpi_c(m)|, \quad t_e := \max_{\substack{u \in C_P(\underline{\Lambda}) \\ e \in \mathbb{Z}_N \setminus \{0\}}} |v_u(e)|. \quad (4.14)$$

Since the one-polymer activities factor over occupied carrier cells, see (3.38), we have

$$|\rho_m(M)| \leq t_m^{|M|}, \quad |\rho_e(E)| \leq t_e^{|E|}. \quad (4.15)$$

The parameters t_m, t_e tend to zero in the code limit (3.6).

For $u \in \mathfrak{C}_s$ and $n \geq 1$, define the number of polymers of size n containing u

$$\mathcal{N}_s(u; n) := \# \{A \in \mathbb{A}_s : u \in \text{car}(A), |A| = n\}. \quad (4.16)$$

We assume that this number grows at most exponentially with increasing polymer size, i.e., there exists a constant $C_s \geq 1$ such that

$$\mathcal{N}_s(u; n) \leq C_s^n \quad \text{for every } u \in \mathfrak{C}_s, n \geq 1. \quad (4.17)$$

This counting hypothesis is automatic on families of carrier adjacency graphs of uniformly bounded degree. Indeed, let Δ_s be the maximal degree of the carrier graph for species s . A connected support of size n containing a fixed carrier cell u has a spanning tree. A depth-first traversal of such a spanning tree produces a closed walk of length $2n - 2$, and hence the number of possible connected supports is at most Δ_s^{2n-2} . Since each occupied carrier cell carries a nonzero coefficient in $\mathbb{Z}_N \setminus \{0\}$, the number of coefficient assignments is at most $(N - 1)^n$. Therefore

$$\mathcal{N}_s(u; n) \leq (N - 1)^n \Delta_s^{2n-2} \leq ((N - 1) \max\{1, \Delta_s^2\})^n. \quad (4.18)$$

Thus one may always take

$$C_s = (N - 1) \max\{1, \Delta_s^2\}. \quad (4.19)$$

A sharper explicit choice follows from the rooted connected-set bound: in a graph of maximum degree Δ_s , the number of connected vertex sets with $k + 1$ vertices containing a fixed root is at most $(e(\Delta_s - 1))^k$; see [40, p. 2]. Applying this to the carrier graph gives

$$\mathcal{N}_s(u; n) \leq (N - 1)^n (e(\Delta_s - 1))^{n-1} \leq ((N - 1) \max\{1, e(\Delta_s - 1)\})^n. \quad (4.20)$$

Hence, whenever $\Delta_s \geq 2$, one may use the improved explicit choice

$$C_s = (N - 1)e(\Delta_s - 1). \quad (4.21)$$

Remark 4.1 (The combinatorial bottleneck). The counting input (4.17) is the weakest point in the present derivation of the low-activity region. The bounds (4.18) and (4.20) effectively count general connected labeled carrier supports, and so do not fully exploit the fact that the polymers are support-connected *closed* cycles. Consequently, the constants C_s in (4.32) and (4.33) are controlled by a purely combinatorial counting problem for rooted connected closed polymers. Any improved bound on $\mathcal{N}_s(u; n)$ incorporating closedness would therefore directly sharpen the rigorous low-activity region, without changing the analytic part of the argument.

4.3 The low-activity criterion

We now combine the activity and counting estimates with the Fernández-Procacci criterion for hard-core polymer gases [21, Theorem 1].

Let $\mathbb{A}$ be a set of hard-core polymers with positive activities $z : \mathbb{A} \rightarrow \mathbb{R}_{\geq 0}$. Write $B \approx A$ to mean that B is incompatible with A and assume $A \approx A$ for all $A \in \mathbb{A}$. For any function $\mu : \mathbb{A} \rightarrow \mathbb{R}_{\geq 0}$ define

$$N_A^*(\mu) := 1 + \sum_{n \geq 1} \frac{1}{n!} \sum_{\substack{B_1, \dots, B_n \in \mathbb{A} \\ B_i \approx A, B_i \sim B_j}} \prod_{i=1}^n \mu(B_i). \quad (4.22)$$

The Fernández-Procacci criterion says that if a positive activity function $\mu : \mathbb{A} \rightarrow \mathbb{R}_{\geq 0}$ exists such that

$$z(A) N_A^*(\mu) \leq \mu(A) \quad \text{for every polymer } A \in \mathbb{A}, \quad (4.23)$$

then the hard-core cluster expansion converges in the large volume limit and the pinned expansion with activity z satisfies⁴

$$z(A) \Pi(A) \leq \mu(A). \quad (4.24)$$

⁴Pinned expansion as in (4.9) but with activity z in place of ρ in (4.6).

For subset polymers with incompatibility given by overlap, the Fernández-Procacci neighborhood partition function admits the sitewise upper bound used below; compare also the polymer-on-a-graph specialization in [21, Section 3.2] and [22, Section 2.3 and Proposition 4.1].

Fix $a_s > 0$ and set

$$\mu_s(A) := |\rho_s(A)| e^{a_s|A|}, \quad A \in \mathbb{A}_s. \quad (4.25)$$

Let

$$S_s(a_s) := \sup_{u \in \mathfrak{C}_s} \sum_{\substack{A \in \mathbb{A}_s \\ u \in \text{car}(A)}} \mu_s(A). \quad (4.26)$$

If

$$S_s(a_s) \leq e^{a_s} - 1, \quad (4.27)$$

then the Fernández-Procacci condition holds for the species- s majorant gas.

Indeed, fix $A \in \mathbb{A}_s$. Every polymer $B \in \mathbb{A}_s$ incompatible with A intersects $\text{car}(A)$. Moreover, in $N_A^*(\mu_s)$, the polymers $B_1, \dots, B_n$ are required to be mutually compatible, hence have disjoint carrier supports. Therefore the contribution of all mutually compatible families $\{B_1, \dots, B_n\}$ incompatible with A is bounded by *independently* choosing, for each carrier cell of A , at most one polymer containing that carrier cell. This gives

$$N_A^*(\mu_s) \leq \prod_{u \in \text{car}(A)} \left(1 + \sum_{\substack{B \in \mathbb{A}_s \\ u \in \text{car}(B)}} |\rho_s(B)| e^{a_s|B|} \right) \leq (1 + S_s(a_s))^{|A|}. \quad (4.28)$$

The first inequality is not an equality because in choosing, for each carrier cell u of A , at most one polymer containing u independently, we are free to choose polymers B, B' containing distinct carrier cells $u, u' \in \text{car}(A)$, $u \in \text{car}(B)$, $u' \in \text{car}(B')$, $u \neq u'$ that are nevertheless incompatible due to $\text{car}(B) \cap \text{car}(B') \neq \emptyset$. If (4.27) holds, then

$$N_A^*(\mu_s) \leq e^{a_s|A|}. \quad (4.29)$$

Consequently,

$$|\rho_s(A)| N_A^*(\mu_s) \leq |\rho_s(A)| e^{a_s|A|} = \mu_s(A), \quad (4.30)$$

which is exactly (4.23).

Using (4.15) and (4.17), we estimate

$$S_s(a_s) \leq \sup_{u \in \mathfrak{C}_s} \sum_{n \geq 1} \sum_{\substack{A \in \mathbb{A}_s \\ u \in \text{car}(A), |A|=n}} t_s^n e^{a_s n} \leq \sum_{n \geq 1} C_s^n (t_s e^{a_s})^n = \frac{C_s t_s e^{a_s}}{1 - C_s t_s e^{a_s}}, \quad (4.31)$$

provided $C_s t_s e^{a_s} < 1$. Thus a sufficient condition for (4.27) is

$$\frac{C_s t_s e^{a_s}}{1 - C_s t_s e^{a_s}} \leq e^{a_s} - 1 \quad \Leftrightarrow \quad C_s t_s \leq e^{-a_s} (1 - e^{-a_s}). \quad (4.32)$$

Definition 4.2 (Rigorous low-activity region). We say that the model lies in the rigorous low-activity region if there exist $a_m, a_e > 0$ such that

$$C_m t_m \leq e^{-a_m} (1 - e^{-a_m}), \quad C_e t_e \leq e^{-a_e} (1 - e^{-a_e}). \quad (4.33)$$

Since $\sup_{a>0} e^{-a} (1 - e^{-a}) = 1/4$, the region is nonempty whenever $C_s t_s < 1/4$, although the parameterized form (4.33) is more useful for tail bounds.

Theorem 4.3 (Low-activity control). *Assume that the physical one-polymer activities satisfy the size bound (4.15), that the polymer counting bound (4.17) holds, and that the corresponding low-activity condition (4.33) is satisfied. In what follows, “convergent” means absolutely convergent uniformly in finite volume, and therefore stable under the large-volume limit.*

For $s \in \{m, e\}$ and $A \in \mathbb{A}_s$, define the unnormalized sector and fixed-background occupation expectations by

$$\langle \mathbf{1}_A \rangle_{\ell_m, \ell_e; q_m, q_e} := \rho^{\ell_m, \ell_e; q_m, q_e}(A) \frac{\partial}{\partial \rho^{\ell_m, \ell_e; q_m, q_e}(A)} \widehat{\mathcal{Z}}_M(\ell_m, \ell_e; q_m, q_e), \quad (4.34)$$

and

$$\langle \mathbf{1}_A \rangle_{q_m, q_e} := \rho_s(A) \frac{\partial}{\partial \rho_s(A)} \mathcal{Z}_M(q_m, q_e). \quad (4.35)$$

Then:

- (i) For every $(\ell_m, \ell_e) \in \widehat{H}_m \times \widehat{H}_e$ and every background pair (q_m, q_e) , the complex sector amplitude $\widehat{\mathcal{Z}}_M(\ell_m, \ell_e; q_m, q_e)$ of (3.48) and the fixed-background partition function $\mathcal{Z}_M(q_m, q_e)$, reconstructed from the sector amplitudes by (3.47), converge.
- (ii) For every $s \in \{m, e\}$, every $A \in \mathbb{A}_s$, every $(\ell_m, \ell_e) \in \widehat{H}_m \times \widehat{H}_e$, and every (q_m, q_e) , the unnormalized occupation expectations $\langle \mathbf{1}_A \rangle_{\ell_m, \ell_e; q_m, q_e}$ and $\langle \mathbf{1}_A \rangle_{q_m, q_e}$ converge. Moreover, there exists a finite constant $C_{M, \underline{\Delta}, s}^{\text{la}}$, independent of A , (ℓ_m, ℓ_e) , and (q_m, q_e) , such that

$$\left| \langle \mathbf{1}_A \rangle_{\ell_m, \ell_e; q_m, q_e} \right| \leq C_{M, \underline{\Delta}, s}^{\text{la}} (t_s e^{a_s})^{|A|}, \quad (4.36)$$

$$\left| \langle \mathbf{1}_A \rangle_{q_m, q_e} \right| \leq C_{M, \underline{\Delta}, s}^{\text{la}} (t_s e^{a_s})^{|A|}. \quad (4.37)$$

Proof. Under (4.15), (4.17), and (4.33), the Fernández-Procacci argument of §4.3 applies to the positive same-species hard-core majorants. In particular, the majorant partition functions are finite, and the corresponding normalized occupation expectations satisfy

$$|\rho_s(A)| \Pi_s^{\text{maj}}(A) \leq |\rho_s(A)| e^{a_s |A|} \leq (t_s e^{a_s})^{|A|}, \quad A \in \mathbb{A}_s. \quad (4.38)$$

Since the complex sector amplitudes are controlled by the majorants through (4.7), this proves the convergence of $\widehat{\mathcal{Z}}_M(\ell_m, \ell_e; q_m, q_e)$ in the low-activity region. The convergence of the fixed-background amplitudes $\mathcal{Z}_M(q_m, q_e)$ follows immediately from (4.8), or equivalently from the finite reconstruction (3.47). This proves (i).

For (ii), differentiate the sector expansion (3.48) with respect to the marked activity $\rho^{\ell_m, \ell_e; q_m, q_e}(A)$. By (4.34),

$$\langle \mathbf{1}_A \rangle_{\ell_m, \ell_e; q_m, q_e} = \rho^{\ell_m, \ell_e; q_m, q_e}(A) \frac{\partial}{\partial \rho^{\ell_m, \ell_e; q_m, q_e}(A)} \widehat{\mathcal{Z}}_M(\ell_m, \ell_e; q_m, q_e). \quad (4.39)$$

Taking absolute values and using the same majorization as in (4.7) yields

$$\left| \langle \mathbf{1}_A \rangle_{\ell_m, \ell_e; q_m, q_e} \right| \leq \mathcal{Z}_{\bar{s}}^{\text{maj}} |\rho_s(A)| \frac{\partial}{\partial |\rho_s(A)|} \mathcal{Z}_s^{\text{maj}}, \quad \bar{m} = e, \bar{e} = m. \quad (4.40)$$

Using (4.10) and then (4.38), we obtain

$$\left| \langle \mathbf{1}_A \rangle_{\ell_m, \ell_e; q_m, q_e} \right| \leq \mathcal{Z}_m^{\text{maj}} \mathcal{Z}_e^{\text{maj}} (t_s e^{a_s})^{|A|}. \quad (4.41)$$

This proves (4.36) at sector level with

$$C_{M, \underline{\Delta}, s}^{\text{la}} := \mathcal{Z}_m^{\text{maj}} \mathcal{Z}_e^{\text{maj}}. \quad (4.42)$$

Similarly,

$$\langle \mathbf{1}_A \rangle_{q_m, q_e} = \rho_s(A) \frac{\partial}{\partial \rho_s(A)} \mathcal{Z}_M(q_m, q_e). \quad (4.43)$$

Differentiating the finite Fourier formula (3.47) and using the sector estimate already proved shows that

$$\left| \langle \mathbf{1}_A \rangle_{q_m, q_e} \right| \leq C_{M, \underline{\Delta}, s}^{\text{la}} (t_s e^{a_s})^{|A|}. \quad (4.44)$$

This gives (4.37). $\square$

Remark 4.4 (Direct fixed-background majorization). The proof of convergence for $\mathcal{Z}_M(q_m, q_e)$ and (4.37) can also be given without passing through the sector amplitudes $\widehat{\mathcal{Z}}_M(\ell_m, \ell_e; q_m, q_e)$. Indeed,

start from (3.40) and take the absolute value. This removes the phase, while the homology constraints are bounded by 1. Hence

$$|\mathcal{Z}_M(q_m, q_e)| \leq \sum_{r,s \geq 0} \frac{1}{r! s!} \sum_{\substack{M_1, \dots, M_r \in \mathbb{M} \\ \text{pairwise disjoint}}} \sum_{\substack{E_1, \dots, E_s \in \mathbb{E} \\ \text{pairwise disjoint}}} \prod_{i=1}^r |\rho_m(M_i)| \prod_{j=1}^s |\rho_e(E_j)| = \mathcal{Z}_m^{\text{maj}} \mathcal{Z}_e^{\text{maj}}. \quad (4.45)$$

This yields the convergence of $\mathcal{Z}_M(q_m, q_e)$ directly, uniformly in the background pair (q_m, q_e) , without invoking (3.48) or (3.47).

The same idea gives the fixed-background occupancy bound. If $A \in \mathbb{A}_s$, differentiate the fixed-background polymer expansion directly with respect to the marked activity $\rho_s(A)$. After taking absolute values, the phase again drops out, the homology constraints are bounded by 1, and one obtains

$$\left| \langle \mathbf{1}_A \rangle_{q_m, q_e} \right| \leq \mathcal{Z}_s^{\text{maj}} |\rho_s(A)| \frac{\partial}{\partial |\rho_s(A)|} \mathcal{Z}_s^{\text{maj}}, \quad \bar{m} = e, \bar{e} = m. \quad (4.46)$$

Using (4.10) and (4.38) then gives (4.37) with the same constant $C_{M, \underline{\Lambda}, s}^{\text{la}} = \mathcal{Z}_m^{\text{maj}} \mathcal{Z}_e^{\text{maj}}$.

4.4 Large polymer suppression

Theorem 4.3 gives an exponential bound on each unnormalized physical one-polymer occupation expectation. Summing that bound over suitable polymer families yields the following two consequences.

Corollary 4.5 (Large polymer suppression). *Assume the hypotheses of Theorem 4.3. Fix a species $s \in \{m, e\}$, and let*

$$\langle \mathbf{1}_A \rangle_{\bullet} \in \left\{ \langle \mathbf{1}_A \rangle_{\ell_m, \ell_e; q_m, q_e}, \langle \mathbf{1}_A \rangle_{q_m, q_e} \right\} \quad (4.47)$$

denote either of the unnormalized physical occupation expectations. Then:

- (i) *The total unnormalized physical occupation weight carried by connected polymers of species s that pass through a fixed carrier cell u and have size at least L is exponentially small in L . Concretely, for every $u \in \mathfrak{C}_s$ and every integer $L \geq 1$,*

$$\sum_{\substack{A \in \mathbb{A}_s \\ u \in \text{car}(A), |A| \geq L}} |\langle \mathbf{1}_A \rangle_{\bullet}| \leq C_{M, \underline{\Lambda}, s}^{\text{la}} \frac{(C_s t_s e^{a_s})^L}{1 - C_s t_s e^{a_s}}. \quad (4.48)$$

- (ii) *In particular, the total unnormalized physical occupation weight carried by connected polymers that are homologically nontrivial, and hence capable of contributing to a change of spacetime topological sector, is exponentially small on the scale of the corresponding spacetime systole. Define*

$$\text{sys}_s(\underline{\Lambda}) := \min \{ |A| : A \in \mathbb{A}_s, [A] \neq 0 \}, \quad s \in \{m, e\}, \quad (4.49)$$

with the convention $+\infty$ if there is no homologically nontrivial connected polymer of species s . Then

$$\sum_{\substack{A \in \mathbb{A}_s \\ [A] \neq 0}} |\langle \mathbf{1}_A \rangle_{\bullet}| \leq C_{M, \underline{\Lambda}, s}^{\text{la}} |\mathfrak{C}_s| \frac{(C_s t_s e^{a_s})^{\text{sys}_s(\underline{\Lambda})}}{1 - C_s t_s e^{a_s}}, \quad (4.50)$$

Proof. By Theorem 4.3,

$$|\langle \mathbf{1}_A \rangle_{\bullet}| \leq C_{M, \underline{\Lambda}, s}^{\text{la}} (t_s e^{a_s})^{|A|} \quad (A \in \mathbb{A}_s). \quad (4.51)$$

Summing (4.51) over all polymers of species s with $u \in \text{car}(A)$ and $|A| \geq L$, and using the counting estimate (4.17), gives

$$\begin{aligned} \sum_{\substack{A \in \mathbb{A}_s \\ u \in \text{car}(A), |A| \geq L}} |\langle \mathbf{1}_A \rangle_{\bullet}| &\leq C_{M, \underline{\Lambda}, s}^{\text{la}} \sum_{n \geq L} \mathcal{N}_s(u; n) (t_s e^{a_s})^n \\ &\leq C_{M, \underline{\Lambda}, s}^{\text{la}} \sum_{n \geq L} C_s^n (t_s e^{a_s})^n = C_{M, \underline{\Lambda}, s}^{\text{la}} \frac{(C_s t_s e^{a_s})^L}{1 - C_s t_s e^{a_s}}, \end{aligned} \quad (4.52)$$

which proves (4.48).

For (4.50), note that every homologically nontrivial polymer of species s has size at least $\text{sys}_s(\underline{\Lambda})$. Applying (4.48) with $L = \text{sys}_s(\underline{\Lambda})$ and then summing over all carrier cells $u \in \mathfrak{C}_s$, gives the stated bound. As before, this overcounts each polymer by its carrier size, which is harmless for an upper bound. $\square$

The bounds of Corollary 4.5 admit a direct physical interpretation. From the statistical-mechanical point of view, in a positive polymer ensemble the normalized occupation expectation of a polymer is the probability that the polymer occurs, and sums over classes of polymers measure the expected number of such excitations. In the present complex theory the sector weights are not positive, so the occupation expectations are amplitudes rather than probabilities. The absolute values in (4.48) and (4.50) therefore remove possible phase cancellations and measure the total absolute contribution carried by the specified family of polymers. The corollary shows that this total contribution is exponentially small for large polymers, and in particular for homologically nontrivial polymers.

From the quantum-mechanical point of view, a connected polymer is an extended Euclidean defect history in the spacetime representation of the thermal trace. A homologically nontrivial polymer is a defect history capable of changing the spacetime topological sector, and hence of contributing to a logical sector-changing process. The bound (4.48) says that long connected defect histories have exponentially small total amplitude weight, while (4.50) says that defect histories capable of changing the topological sector are exponentially suppressed on the scale of the spacetime systole. Thus, within the low-activity region, the total absolute weight carried by dangerous large or sector-changing thermal histories is perturbatively small.

5 Kramers-Wannier duality

We now derive the exact duality transformation of the spacetime theory. This is the finite- $\mathbb{Z}_N$, higher-form analogue of the Kramers-Wannier/Wegner duality [6, 7, 24, 25]; for a modern algebraic viewpoint on exact lattice dualities, see [29]. Historically, one of the key lessons of the Kramers-Wannier program is that duality is often revealed by Fourier transform. In the classical lattice setting, the low-temperature/high-temperature transform can be recast as a local Fourier transform of Boltzmann weights, and in Abelian models this became the standard route to dual partition functions [6, 7, 24]. More recently, the same Fourier-theoretic structure has been reinterpreted from a higher-dimensional and topological viewpoint: for finite gauge theories and generalized Ising-type systems, Kramers-Wannier duality can be understood as the boundary manifestation of a duality in one dimension higher, with the Fourier transform acting on local weights or background sectors [30, 31].

Our derivation follows this general philosophy, applying the Fourier transform directly to the exact fixed-background amplitudes of §3. The natural observables are the fixed-background partition functions

$$\mathcal{Z}_M(q_m, q_e), \quad q_m \in Z_{d-P}(\underline{\Lambda}^\vee), \quad q_e \in Z_P(\underline{\Lambda}). \quad (5.1)$$

The cleanest route is to work first in the fixed-background basis, perform a cellwise finite Fourier transform of the local $(P+1)$ -cell weights, and reinterpret the resulting auxiliary variables as a dual $(d-P)$ -form field on $\underline{\Lambda}^\vee$. This exchanges magnetic and electric background data, simultaneously exchanges the P -form theory on $\underline{\Lambda}$ with a $(d-P)$ -form theory on $\underline{\Lambda}^\vee$, and turns the local Fourier transform of weights into the exact dual coupling map. Throughout this section N is any integer ≥ 2 and,

$$W_c : \mathbb{Z}_N \rightarrow \mathbb{C} \quad \text{for } c \in C_{P+1}(\underline{\Lambda}), \quad V_u : \mathbb{Z}_N \rightarrow \mathbb{C} \quad \text{for } u \in C_P(\underline{\Lambda}) \quad (5.2)$$

are arbitrary local spacetime weights. At the end we specialize to the explicit Trotter weights (2.21), (2.22). We use the Fourier-transform conventions of Appendix A.

5.1 Duality in the fixed-charge basis

We begin with the unnormalized fixed-charge partition function (3.3), but for the present discussion we make the weight dependence explicit:

$$Z_M(q_m, q_e; W, V) := \sum_{\phi \in \Omega^P(\underline{\Lambda})} \chi \left(\int_{q_e} \phi \right) \prod_{c \in C_{P+1}(\underline{\Lambda})} W_c((D\phi + \vartheta_{\underline{\Lambda}}(q_m))_c) \prod_{u \in C_P(\underline{\Lambda})} V_u(\phi_u). \quad (5.3)$$

5.1.1 Fourier expansion of the $(P+1)$ -cell weights

Replace W_c in (5.3) by its Fourier series. This introduces an auxiliary $(P+1)$ -cochain $\lambda \in \Omega^{P+1}(\underline{\Lambda})$ and gives

$$\begin{aligned} Z_M(q_m, q_e; W, V) &= N^{-|C_{P+1}(\underline{\Lambda})|/2} \sum_{\phi, \lambda} \chi \left(\int_{q_e} \phi \right) \left(\prod_{c \in C_{P+1}(\underline{\Lambda})} \widehat{W}_c(\lambda_c) \chi_{\lambda_c}((\vartheta_{\underline{\Lambda}}(q_m))_c) \right) \\ &\quad \times \prod_{c \in C_{P+1}(\underline{\Lambda})} \chi_{\lambda_c}((D\phi)_c) \prod_{u \in C_P(\underline{\Lambda})} V_u(\phi_u). \end{aligned} \quad (5.4)$$

Next move the differential D from ϕ onto λ . By the adjointness of D and $D^\top$ (see (B.28)),

$$\sum_{c \in C_{P+1}(\underline{\Lambda})} \lambda_c (D\phi)_c = \sum_{u \in C_P(\underline{\Lambda})} (D^\top \lambda)_u \phi_u. \quad (5.5)$$

Hence,⁵

$$\chi \left(\int_{q_e} \phi \right) \prod_{c \in C_{P+1}(\underline{\Lambda})} \chi_{\lambda_c}((D\phi)_c) = \prod_{u \in C_P(\underline{\Lambda})} \chi_{(\mathcal{P}_{\underline{\Lambda}}^{-1}(q_e) + D^\top \lambda)_u}(\phi_u), \quad (5.6)$$

and the ϕ -dependence factorizes cellwise:

$$\begin{aligned} Z_M(q_m, q_e; W, V) &= N^{-|C_{P+1}(\underline{\Lambda})|/2} \sum_{\lambda \in \Omega^{P+1}(\underline{\Lambda})} \left(\prod_{c \in C_{P+1}(\underline{\Lambda})} \widehat{W}_c(\lambda_c) \chi_{\lambda_c}((\vartheta_{\underline{\Lambda}}(q_m))_c) \right) \\ &\quad \times \prod_{u \in C_P(\underline{\Lambda})} \left[\sum_{x \in \mathbb{Z}_N} V_u(x) \chi_{(\mathcal{P}_{\underline{\Lambda}}^{-1}(q_e) + D^\top \lambda)_u}(x) \right]. \end{aligned} \quad (5.7)$$

Using the inverse Fourier transform,

$$\sum_{x \in \mathbb{Z}_N} V_u(x) \chi_k(x) = \sqrt{N} \widehat{V}_u(-k), \quad (5.8)$$

we obtain

$$\begin{aligned} \frac{Z_M(q_m, q_e; W, V)}{N^{|C_P(\underline{\Lambda})|/2}} &= \frac{1}{N^{|C_{\check{P}}(\underline{\Lambda}^\vee)|/2}} \sum_{\lambda \in \Omega^{P+1}(\underline{\Lambda})} \left(\prod_{c \in C_{P+1}(\underline{\Lambda})} \widehat{W}_c(\lambda_c) \chi_{\lambda_c}((\vartheta_{\underline{\Lambda}}(q_m))_c) \right) \\ &\quad \times \left(\prod_{u \in C_P(\underline{\Lambda})} \widehat{V}_u \left(-(D^\top \lambda + \mathcal{P}_{\underline{\Lambda}}^{-1}(q_e))_u \right) \right), \end{aligned} \quad (5.9)$$

with

$$\check{P} := d - P. \quad (5.10)$$

5.1.2 Reinterpretation on the dual spacetime lattice

Define the dual field

$$\check{\phi} \in \Omega^{\check{P}}(\underline{\Lambda}^\vee) \quad (5.11)$$

by transporting λ across the primal-dual correspondence:

$$\check{\phi}_{c^\vee} := \lambda_{\theta_{\underline{\Lambda}}(c^\vee)}, \quad c^\vee \in C_{\check{P}}(\underline{\Lambda}^\vee). \quad (5.12)$$

The intertwining of $D^\top$ on the primal side with $D^\vee$ on the dual side (see Prop. B.1(iii)) gives

$$(D^\top \lambda)_{\theta_{\underline{\Lambda}}(u^\vee)} = (D^\vee \check{\phi})_{u^\vee}, \quad u^\vee \in C_{\check{P}+1}(\underline{\Lambda}^\vee). \quad (5.13)$$

⁵We use the isomorphism (B.24) strictly for typecasting: change the P -cycle q_e to a P -cocycle $\mathcal{P}_{\underline{\Lambda}}^{-1}(q_e)$. They have the same coefficients, concretely, $\mathcal{P}_{\underline{\Lambda}}^{-1}(q_e)_u = (q_e)_u$.

The magnetic charge term becomes a Wilson factor of the dual theory:

$$\prod_{c \in C_{P+1}(\underline{\Lambda})} \chi_{\lambda_c}((\vartheta_{\underline{\Lambda}}(q_m))_c) = \prod_{c^\vee \in C_{\check{P}}(\underline{\Lambda}^\vee)} \chi_{\check{\phi}_{c^\vee}}((q_m)_{c^\vee}) = \chi\left(\int_{q_m} \check{\phi}\right). \quad (5.14)$$

And the electric charge becomes a defect cocycle of the dual theory. Define

$$q_e^\vee := \vartheta_{\underline{\Lambda}^\vee}(q_e) \in Z^{\check{P}+1}(\underline{\Lambda}^\vee), \quad (5.15)$$

which is a cocycle since ϑ is an isomorphism (Corr. B.2). Then

$$(D^\top \lambda + \mathcal{P}_{\underline{\Lambda}}^{-1}(q_e))_{\theta_{\underline{\Lambda}}(u^\vee)} = (D^\vee \check{\phi} + q_e^\vee)_{u^\vee}. \quad (5.16)$$

The natural dual local weights are therefore (using the map $\theta^\sharp$ from (B.32) and defining for any function f , $f^-(x) := f(-x)$)

$$\check{W} := \theta_{\underline{\Lambda}^\vee}^\sharp(\widehat{V}^-), \quad \check{V} := \theta_{\underline{\Lambda}^\vee}^\sharp(\widehat{W}), \quad (5.17)$$

or explicitly,

$$\check{W}_{u^\vee} := \widehat{V}_{\theta_{\underline{\Lambda}}(u^\vee)}^-, \quad u^\vee \in C_{\check{P}+1}(\underline{\Lambda}^\vee), \quad (5.18)$$

and

$$\check{V}_{c^\vee} := \widehat{W}_{\theta_{\underline{\Lambda}}(c^\vee)}, \quad c^\vee \in C_{\check{P}}(\underline{\Lambda}^\vee). \quad (5.19)$$

This yields the dual $\check{P}$ -form partition function on $\underline{\Lambda}^\vee$:

$$\begin{aligned} Z_M^\vee(q_e, q_m; \check{W}, \check{V}) &:= \sum_{\check{\phi} \in \Omega^{\check{P}}(\underline{\Lambda}^\vee)} \chi\left(\int_{q_m} \check{\phi}\right) \prod_{u^\vee \in C_{\check{P}+1}(\underline{\Lambda}^\vee)} \check{W}_{u^\vee} \left((D^\vee \check{\phi} + q_e^\vee)_{u^\vee}\right) \\ &\quad \times \prod_{c^\vee \in C_{\check{P}}(\underline{\Lambda}^\vee)} \check{V}_{c^\vee}(\check{\phi}_{c^\vee}). \end{aligned} \quad (5.20)$$

Proposition 5.1 (Duality with background charges). *With $\check{P} = d - P$, the background-dependent partition functions have the following duality properties:*

(i) *The unnormalized fixed-charge partition functions obey*

$$\frac{Z_M(q_m, q_e; W, V)}{N^{|C_P(\underline{\Lambda})|/2}} = \frac{Z_M^\vee(q_e, q_m; \check{W}, \check{V})}{N^{|C_{\check{P}}(\underline{\Lambda}^\vee)|/2}}. \quad (5.21)$$

In particular, duality exchanges the electric and magnetic fixed charges and simultaneously exchanges the P -form theory on $\underline{\Lambda}$ with the $\check{P}$ -form theory on $\underline{\Lambda}^\vee$.

(ii) *Let $\mathcal{Z}_M(q_m, q_e; W, V)$ and $\mathcal{Z}_M^\vee(q_e, q_m; \check{W}, \check{V})$ denote the normalized fixed-charge amplitudes defined as in (3.28) for the primal and dual theories respectively. Then*

$$\frac{|Z^P(\underline{\Lambda})|}{N^{|C_P(\underline{\Lambda})|}} \mathcal{Z}_M(q_m, q_e; W, V) = \frac{|Z^{\check{P}}(\underline{\Lambda}^\vee)|}{N^{|C_{\check{P}}(\underline{\Lambda}^\vee)|}} \mathcal{Z}_M^\vee(q_e, q_m; \check{W}, \check{V}). \quad (5.22)$$

Proof. Item (i) follows by substituting (5.14), (5.16), (5.18), and (5.19) into (5.9).

For item (ii), let $\mathcal{N}_M^\vee$ be the dual analogue of (3.20), namely the normalization obtained by replacing $(\underline{\Lambda}, P, W, V)$ with $(\underline{\Lambda}^\vee, \check{P}, \check{W}, \check{V})$. Using (5.18), (5.19), and the symmetric Fourier convention, we have

$$\check{W}_{u^\vee}(0) = \widehat{V}_{\theta_{\underline{\Lambda}}(u^\vee)}(0), \quad \widehat{V}_{c^\vee}(0) = W_{\theta_{\underline{\Lambda}}(c^\vee)}(0), \quad (5.23)$$

so that

$$\mathcal{N}_M = N^{\frac{-|C_P(\underline{\Lambda})| + |C_{\check{P}}(\underline{\Lambda}^\vee)|}{2}} \mathcal{N}_M^\vee. \quad (5.24)$$

Now dividing both sides of (5.21) by $\mathcal{N}_M$ gives (5.22). $\square$

5.2 Specialization to the Trotter weights

We now specialize (5.18) and (5.19) to the explicit local weights (2.22) and (2.21). The key point is that, with the present Fourier conventions, the transformed weights again have the same functional form as the original anisotropic Trotter weights, up to local field-independent factors. This makes the dual coupling map transparent.

Dual horizontal $(\check{P} + 1)$ -cell weights. Let

$$c^\vee \in C_{\check{P}+1}(\Lambda^\vee), \quad a := \theta_\Lambda(c^\vee) \in C_{P-1}(\Lambda). \quad (5.25)$$

Then the horizontal dual cell $c^\vee(i)$ corresponds to the vertical primal cell $\underline{a}(i)$, so

$$\check{W}_{c^\vee(i)}^\parallel(m) = \widehat{V}_{\underline{a}(i)}^\perp(-m). \quad (5.26)$$

Using (2.22a),

$$\begin{aligned} \widehat{V}_{\underline{a}(i)}^\perp(-m) &= \frac{1}{\sqrt{N}} \sum_{x \in \mathbb{Z}_N} \left(\delta_N(x) + \frac{e^{\beta J_a/M} - 1}{N} \right) \chi_{-m}(-x) \\ &= \frac{1}{\sqrt{N}} \left(1 + \left(e^{\beta J_a/M} - 1 \right) \delta_N(m) \right) \\ &= \frac{1}{\sqrt{N}} \exp \left(\frac{\beta J_a}{M} \delta_N(m) \right). \end{aligned} \quad (5.27)$$

Thus, up to the field-independent factor $N^{-1/2}$, the dual horizontal weight is again of the form (2.21a). We therefore define

$$\check{K}_{c^\vee} := J_{\theta_\Lambda(c^\vee)}. \quad (5.28)$$

Dual vertical $\check{P}$ -cell weights. Let

$$a^\vee \in C_{\check{P}-1}(\Lambda^\vee), \quad c := \theta_\Lambda(a^\vee) \in C_{P+1}(\Lambda). \quad (5.29)$$

Then the vertical dual cell $\underline{a}^\vee(i)$ corresponds to the horizontal primal cell $c(i)$, so

$$\check{V}_{\underline{a}^\vee(i)}^\perp(x) = \widehat{W}_{c(i)}^\parallel(x). \quad (5.30)$$

Using (2.21a),

$$\begin{aligned} \widehat{W}_{c(i)}^\parallel(x) &= \frac{1}{\sqrt{N}} \sum_{m \in \mathbb{Z}_N} \left(1 + \left(e^{\beta K_c/M} - 1 \right) \delta_N(m) \right) \chi_x(-m) \\ &= \sqrt{N} \left(\delta_N(x) + \frac{e^{\beta K_c/M} - 1}{N} \right). \end{aligned} \quad (5.31)$$

Thus, up to the factor $\sqrt{N}$, the dual vertical weight is again of the form (2.22a), and we define

$$\check{J}_{a^\vee} := K_{\theta_\Lambda(a^\vee)}. \quad (5.32)$$

Dual vertical $(\check{P} + 1)$ -cell source weights. Let

$$u^\vee \in C_{\check{P}}(\Lambda^\vee), \quad u := \theta_\Lambda(u^\vee) \in C_P(\Lambda). \quad (5.33)$$

Then the vertical dual $(\check{P} + 1)$ -cell $\underline{u}^\vee(i)$ corresponds to the horizontal primal P -cell $u(i)$, so

$$\check{W}_{\underline{u}^\vee(i)}^\perp(m) = \widehat{V}_{u(i)}^\parallel(-m). \quad (5.34)$$

Using (2.21b),

$$\begin{aligned}\widehat{V}_{u(i)}^{\parallel}(-m) &= \frac{1}{\sqrt{N}} \sum_{x \in \mathbb{Z}_N} \exp \left(\frac{\beta}{M} \sum_{n \in \mathbb{Z}_N} h_u^{(n)} \omega^{nx} \right) \omega^{mx} \\ &= \frac{\sqrt{N}}{N} \sum_{x \in \mathbb{Z}_N} \exp \left(\frac{\beta}{M} \sum_{n \in \mathbb{Z}_N} h_u^{(n)} \omega^{nx} \right) \omega^{mx}.\end{aligned}\quad (5.35)$$

Comparing with the standard vertical weight for a $\check{P}$ -form theory on $\Lambda^\vee$, one finds, after the change of variable $x = (-1)^{\check{P}} j$, that the functional form agrees up to the field-independent factor $\sqrt{N}$, provided

$$\check{g}_{u^\vee}^{(n)} = h_u^{((-1)^{\check{P}} n)} = h_u^{((-1)^{d-P} n)}. \quad (5.36)$$

Dual horizontal $\check{P}$ -cell source weights. Let

$$u^\vee \in C_{\check{P}}(\Lambda^\vee), \quad u := \theta_\Lambda(u^\vee) \in C_P(\Lambda). \quad (5.37)$$

Then the horizontal dual $\check{P}$ -cell $u^\vee(i)$ corresponds to the vertical primal $(P+1)$ -cell $\underline{u}(i)$, so

$$\check{V}_{u^\vee(i)}^{\parallel}(x) = \widehat{W}_{\underline{u}(i)}^\perp(x). \quad (5.38)$$

Using (2.22b),

$$\begin{aligned}\widehat{W}_{\underline{u}(i)}^\perp(x) &= \frac{1}{\sqrt{N}} \sum_{m \in \mathbb{Z}_N} \left[\frac{1}{N} \sum_{j \in \mathbb{Z}_N} \exp \left(\frac{\beta}{M} \sum_{n \in \mathbb{Z}_N} g_u^{(n)} \omega^{nj} \right) \omega^{(-1)^P j m} \right] \omega^{-xm} \\ &= \frac{1}{\sqrt{N}} \exp \left(\frac{\beta}{M} \sum_{n \in \mathbb{Z}_N} g_u^{(n)} \omega^{n((-1)^P x)} \right).\end{aligned}\quad (5.39)$$

Thus, up to the factor $N^{-1/2}$, the dual horizontal weight is again of the form (2.21b), with

$$\check{h}_{u^\vee}^{(n)} = g_u^{((-1)^P n)}. \quad (5.40)$$

Collecting (5.28), (5.32), (5.36), and (5.40), we obtain the dual coupling map

$$\check{K}_{c^\vee} = J_{\theta_\Lambda(c^\vee)}, \quad \check{J}_{a^\vee} = K_{\theta_\Lambda(a^\vee)}, \quad (5.41)$$

and

$$\check{g}_{u^\vee}^{(n)} = h_{\theta_\Lambda(u^\vee)}^{((-1)^{d-P} n)}, \quad \check{h}_{u^\vee}^{(n)} = g_{\theta_\Lambda(u^\vee)}^{((-1)^P n)}. \quad (5.42)$$

5.2.1 Cancellation of the local factors

The transformed local weights (5.27), (5.31), (5.35), and (5.39) differ from the canonical Trotter weights only by local field-independent factors $N^{\pm 1/2}$. These local factors exactly cancel the factors of N in (5.21), as we now show. Comparing the Fourier transformed local factors with the original local factors (2.21) and (2.22) we obtain one factor $N^{-1/2}$ for each dual horizontal $(\check{P}+1)$ -cell, one factor $N^{1/2}$ for each dual vertical $(\check{P}+1)$ -cell, one factor $N^{1/2}$ for each dual vertical $\check{P}$ -cell, and one factor $N^{-1/2}$ for each dual horizontal $\check{P}$ -cell. Their total product is therefore

$$\begin{aligned}\mathfrak{F}_{P,\underline{\Lambda}} &:= \prod_{i \in \mathbb{Z}/M\mathbb{Z}} \prod_{a \in C_{P-1}(\Lambda)} N^{-1/2} \prod_{c \in C_{P+1}(\Lambda)} N^{1/2} \prod_{b \in C_P(\Lambda)} N^{1/2} \prod_{b \in C_P(\Lambda)} N^{-1/2} \\ &= N^{\frac{M}{2}(-|C_{P-1}(\Lambda)| + |C_{P+1}(\Lambda)|)} = N^{\frac{|C_{P+1}(\underline{\Lambda})| - |C_P(\underline{\Lambda})|}{2}} = N^{\frac{|C_{\check{P}}(\underline{\Lambda}^\vee)| - |C_P(\underline{\Lambda})|}{2}}\end{aligned}\quad (5.43)$$

Thus the transformed dual partition function differs from the canonical dual Trotter partition function only by the factor $\mathfrak{F}_{P,\underline{\Lambda}}$, and this cancels exactly against the normalization factors in (5.21). It is convenient to record the Trotter amplitudes explicitly:

$$Z_{P,M}(q_m, q_e; J, K, g, h) := Z_M(q_m, q_e; W[J, K, g, h], V[J, K, g, h]). \quad (5.44)$$

Theorem 5.2 (Exact Kramers-Wannier duality for the Trotter spacetime theory). *For the explicit Trotter weights (2.21), (2.22), the fixed-charge duality is normalization-free:*

$$Z_{P,M}(q_m, q_e; J, K, g, h) = Z_{\tilde{P},M}^\vee(q_e, q_m; \check{J}, \check{K}, \check{g}, \check{h}), \quad (5.45)$$

where the dual couplings are given by (5.41) and (5.42).

Proof. Equation (5.43) shows that the local Fourier prefactors produced in the specialization to the Trotter weights cancel exactly against the normalization factors in (5.21), yielding (5.45). $\square$

5.3 Duality of the quantum code

The couplings J, K, g, h are precisely the couplings entering the quantum Hamiltonian of §2: J and K multiply the X - and Z -type stabilizer projectors $\hat{\mathcal{A}}_a$ and $\hat{\mathcal{B}}_c$ respectively; g and h couple to local X - and Z -type source terms respectively. The dual coupling map (5.41)-(5.42) therefore exchanges X -type and Z -type operators under the primal-dual correspondence.

On the source-free stabilizer part of the Hamiltonian, this is already visible from (5.41): the coefficient of a dual X -type stabilizer comes from a primal Z -type stabilizer, and the coefficient of a dual Z -type stabilizer comes from a primal X -type stabilizer. Thus the P -form code on Λ is dual to the $(d - P)$ -form code on $\Lambda^\vee$, with electric and magnetic stabilizers interchanged.

The same statement extends to the local Weyl operators themselves. If $u^\vee \in C_{\tilde{P}}(\Lambda^\vee)$ and $u = \theta_\Lambda(u^\vee) \in C_P(\Lambda)$, then the source-mode map is

$$\hat{Z}_u^n \longleftrightarrow \check{X}_{u^\vee}^{(-1)^{d-P}n}, \quad \hat{X}_u^n \longleftrightarrow \check{Z}_{u^\vee}^{(-1)^Pn}. \quad (5.46)$$

Apart from these orientation signs, duality simply exchanges X and Z .

At the level of logical operators, the exchange is equally concrete. Let

$$\nu \in Z_P(\Lambda), \quad \mu^\vee \in Z_{d-P}(\Lambda^\vee) \quad (5.47)$$

label the spatial Wilson and 't Hooft operators $\hat{\mathcal{W}}_\nu$ and $\hat{\mathcal{T}}_{\mu^\vee}$. Under the primal-dual correspondence,

$$\hat{\mathcal{W}}_\nu \longleftrightarrow \check{\mathcal{T}}_\nu, \quad \hat{\mathcal{T}}_{\mu^\vee} \longleftrightarrow \check{\mathcal{W}}_{\mu^\vee}. \quad (5.48)$$

Thus Kramers-Wannier duality exchanges the electric and magnetic logical operator algebras of the code.

If the spatial lattice is self-dual, so that Λ is identified with $\Lambda^\vee$, the duality acts internally on the family of quantum codes. The self-dual locus is then the fixed-point set of the coupling map:

$$P = d - P, \quad J_a = K_{\theta_\Lambda^{-1}(a)}, \quad g_u^{(n)} = h_{\theta_\Lambda^{-1}(u)}^{((-1)^{d-P}n)}. \quad (5.49)$$

In the homogeneous case this reduces to

$$P = \frac{d}{2}, \quad J = K, \quad g^{(n)} = h^{((-1)^Pn)}. \quad (5.50)$$

If, in addition, the source coefficients are parity-symmetric,

$$g^{(n)} = g^{(-n)}, \quad h^{(n)} = h^{(-n)}, \quad (5.51)$$

the sign in (5.50) becomes irrelevant, and the self-dual locus takes the simpler form

$$P = \frac{d}{2}, \quad J = K, \quad g = h. \quad (5.52)$$

On this locus the quantum Hamiltonian is invariant under the combined operation of primal-dual identification and exchange of X -type and Z -type operators. In particular, the electric and magnetic logical sectors are related by an exact symmetry of the code.

This completes the derivation of the exact higher-form Kramers-Wannier symmetry of the spacetime theory. In the fixed-charge basis, duality exchanges electric and magnetic backgrounds. For the explicit Trotter spacetime theory, the fixed-charge duality acts without any residual normalization factor. At

the level of the quantum code, it interchanges X -type and Z -type stabilizers, local Weyl operators, and Wilson/'t Hooft logical observables under the primal-dual correspondence. The importance of this symmetry is twofold. First, it shows that the full spacetime theory space carries an exact $\mathbb{Z}_2$ -involution. Second, it identifies special slices on which one expects sharper structure than is visible in the generic perturbative analysis. The next section studies precisely such slices: the exact source-free gauge theory-subspace and its plaquette random-cluster companion, where stronger nonperturbative results can be imported.

6 Gauge/PRCM specializations

The background-dependent sectorwise polymer gas of Sections 3-4 allows arbitrary local weights, complex electric-magnetic linking phases, and no positivity or monotonicity assumptions. Its rigorous control is therefore perturbative. There is, however, a distinguished exact positive subspace of the full spacetime weight space on which the theory reduces to a source-free $\mathbb{Z}_N$ lattice gauge model. This places the present construction in the broader landscape of lattice gauge theory, duality, and phase structure initiated by Wegner and developed further in [6, 32, 33].

For prime N , this gauge theory is exactly coupled to the plaquette random-cluster model (PRCM). On closed lattices with nontrivial ambient homology, the most relevant sharp result presently available is the middle-dimensional self-dual toric transition of Duncan-Schweinhart [37].

6.1 The exact gauge/PRCM specialization

Let X be a finite oriented cell complex of dimension n . For arbitrary local weights $W_c : \mathbb{Z}_N \rightarrow \mathbb{C}$ on $(P+1)$ -cells and $V_u : \mathbb{Z}_N \rightarrow \mathbb{C}$ on P -cells, define

$$\mathcal{Z}_X(W, V) := \sum_{\phi \in \Omega^P(X)} \prod_{c \in C_{P+1}(X)} W_c((d_X \phi)_c) \prod_{u \in C_P(X)} V_u(\phi_u). \quad (6.1)$$

In particular, for $X = \underline{\Lambda}$ and $q_m = q_e = 0$, this is the neutral-background spacetime partition function (2.23).

The exact source-free gauge theory-subspace is obtained by removing the P -cell weights and retaining only a flatness weight on $(P+1)$ -cells. Writing $\beta_g \geq 0$ for the gauge coupling, set

$$V_u(x) = 1 \quad (u \in C_P(X)), \quad W_c(x) = 1 + (e^{\beta_g} - 1) \delta_N(x) \quad (c \in C_{P+1}(X)). \quad (6.2)$$

Then (6.1) becomes

$$\begin{aligned} \mathcal{Z}_X^{\text{gauge}}(\beta_g) &:= \sum_{\phi \in \Omega^P(X)} \prod_{c \in C_{P+1}(X)} \left[1 + (e^{\beta_g} - 1) \delta_N((d_X \phi)_c) \right] \\ &= \sum_{\phi \in \Omega^P(X)} \exp \left(\beta_g \sum_{c \in C_{P+1}(X)} \delta_N((d_X \phi)_c) \right). \end{aligned} \quad (6.3)$$

This is the P -form N -state Potts lattice gauge theory on X , written in additive $\mathbb{Z}_N$ notation [6, 32, 33, 36–38]. In the absence of sources the action is manifestly invariant under a gauge transformation: $\phi \mapsto \phi + d_X \psi$ for $\psi \in \Omega^{P-1}(X)$.

6.1.1 Potts lattice gauge theory from the Trotter weights

It is useful to locate this gauge theory-subspace precisely inside the anisotropic Trotter family of §2. Fix the Trotter number M and take $X = \underline{\Lambda}$. The horizontal P -cell weights (2.21b) become trivial exactly when

$$h_b^{(n)} = 0 \quad \text{for all } b \in C_P(\Lambda), \quad n \in \mathbb{Z}_N, \quad (6.4)$$

so that $V_{b(i)}^{\parallel}(x) \equiv 1$. The vertical P -cell weights are (2.22a),

$$V_{\underline{a}(i)}^{\perp}(x) = \delta_N(x) + \frac{e^{\beta J_a/M} - 1}{N}. \quad (6.5)$$

After dividing by the field-independent zero-mode factor $V_{\underline{a}(i)}^\perp(0)$, one has

$$\lim_{J_a \rightarrow \infty} \frac{V_{\underline{a}(i)}^\perp(x)}{V_{\underline{a}(i)}^\perp(0)} \rightarrow 1 \quad \text{uniformly in } x \in \mathbb{Z}_N. \quad (6.6)$$

Thus the P -cell weights are removed by the exact condition $h = 0$ together with the strong-coupling limit $J \rightarrow \infty$.

For the $(P+1)$ -cell weights, the horizontal part (2.21a) already has gauge form:

$$W_{c(i)}^\parallel(x) = 1 + \left(e^{\beta K_c/M} - 1\right) \delta_N(x). \quad (6.7)$$

Hence a prescribed horizontal gauge coupling $\beta_{c(i)}^\parallel$ is realized by choosing

$$K_c = \frac{M}{\beta} \beta_{c(i)}^\parallel. \quad (6.8)$$

The vertical $(P+1)$ -cell weights are (2.22b),

$$W_{\underline{b}(i)}^\perp(x) = \frac{1}{N} \sum_{j \in \mathbb{Z}_N} \exp\left(\frac{\beta}{M} \sum_{n \in \mathbb{Z}_N} g_b^{(n)} \omega^{nj}\right) \omega^{(-1)^P jx}. \quad (6.9)$$

To realize a prescribed vertical gauge coupling $\beta_{\underline{b}(i)}^\perp$, define

$$F_{\underline{b}(i)}(j) := \left(e^{\beta_{\underline{b}(i)}^\perp} - 1\right) + N \delta_N(j), \quad j \in \mathbb{Z}_N, \quad (6.10)$$

and

$$\Gamma_{\underline{b}(i)}(j) := \log F_{\underline{b}(i)}(j) - \frac{1}{N} \sum_{k \in \mathbb{Z}_N} \log F_{\underline{b}(i)}(k). \quad (6.11)$$

Now define g to essentially be the Fourier transform of Γ :

$$g_b^{(n)} := \frac{M}{\beta N} \sum_{j \in \mathbb{Z}_N} \Gamma_{\underline{b}(i)}(j) \omega^{-nj}. \quad (6.12)$$

Then $g_b^{(0)} = 0$, and discrete Fourier inversion gives

$$\exp\left(\frac{\beta}{M} \sum_{n \in \mathbb{Z}_N} g_b^{(n)} \omega^{nj}\right) \propto F_{\underline{b}(i)}(j), \quad (6.13)$$

with proportionality factor independent of j . Consequently,

$$W_{\underline{b}(i)}^\perp(x) \propto \frac{1}{N} \sum_{j \in \mathbb{Z}_N} F_{\underline{b}(i)}(j) \omega^{(-1)^P jx} = 1 + \left(e^{\beta_{\underline{b}(i)}^\perp} - 1\right) \delta_N(x). \quad (6.14)$$

Therefore, after the zero-mode normalization of the P -cell weights and the limit $J \rightarrow \infty$, the Trotter family reduces to an anisotropic source-free gauge theory on $\underline{\Lambda}$ with horizontal couplings $\beta_{c(i)}^\parallel$ and vertical couplings $\beta_{\underline{b}(i)}^\perp$. The isotropic gauge theory-subspace is the further specialization

$$\beta_{c(i)}^\parallel = \beta_{\underline{b}(i)}^\perp = \beta_g. \quad (6.15)$$

Strong coupling limit of the gauge theory as the code limit. This also clarifies the relation with the code limit (3.6). The source-free gauge theory is an exact finite-coupling subspace of the general spacetime weight space (6.1). Inside the explicit anisotropic Trotter family, the same gauge theory is reached after the zero-mode normalization of the P -cell weights, the exact condition $h = 0$, the limiting condition $J \rightarrow \infty$, and the choice of K and g in (6.8) and (6.12). One then reaches the code limit by sending the resulting gauge couplings to infinity:

$$\beta_{c(i)}^\parallel \rightarrow \infty, \quad \beta_{\underline{b}(i)}^\perp \rightarrow \infty. \quad (6.16)$$

In that further limit,

$$\frac{W_c(x)}{W_c(0)} \rightarrow \delta_N(x), \quad \frac{V_u(x)}{V_u(0)} \rightarrow 1, \quad (6.17)$$

which is precisely the flat-field regime of §3.1.

6.1.2 Gauge theory coupled to PRCM

The source-free Potts gauge theory (6.3) can be coupled to the plaquette random-cluster model (PRCM) in the standard Edwards-Sokal sense [35]: both measures arise as marginals of a single joint measure. This is exactly the coupling derived in [37, Prop. 20], we record the joint measure and its two marginals in the present notation.

Assume N is prime, so that $\mathbb{Z}_N = \mathbb{F}_N$, and remain on the exact gauge theory-subspace (6.2). Introduce the PRCM probability parameter

$$p := 1 - e^{-\beta_g}. \quad (6.18)$$

Let $\omega(c) \in \{0, 1\}$ be an auxiliary occupation variable on each $(P+1)$ -cell $c \in C_{P+1}(X)$. We then define a joint measure on

$$(\phi, \omega) \in \Omega^P(X) \times \{0, 1\}^{C_{P+1}(X)} \quad (6.19)$$

by

$$\kappa(\phi, \omega) \propto \prod_{c \in C_{P+1}(X)} \left[(1-p) \delta_N(\omega(c)) + p \delta_N(\omega(c) - 1) \delta_N((d_X \phi)_c) \right]. \quad (6.20)$$

Summing over the occupation variables gives

$$\begin{aligned} \sum_{\omega} \kappa(\phi, \omega) &\propto \prod_{c \in C_{P+1}(X)} \left[(1-p) + p \delta_N((d_X \phi)_c) \right] \\ &= (1-p)^{|C_{P+1}(X)|} \prod_{c \in C_{P+1}(X)} \left[1 + \frac{p}{1-p} \delta_N((d_X \phi)_c) \right]. \end{aligned} \quad (6.21)$$

Since $p/(1-p) = e^{\beta_g} - 1$, the ϕ -marginal is, up to the field-independent prefactor $(1-p)^{|C_{P+1}(X)|}$, exactly the source-free Potts gauge measure (6.3).

Conversely, for fixed ω , let $Y = Y(\omega)$ be the subcomplex of X obtained by taking the full P -skeleton of X and adjoining precisely those $(P+1)$ -cells c for which $\omega(c) = 1$. Then $C_{P+1}(Y) = \{c \in C_{P+1}(X) : \omega(c) = 1\}$, and summing over the gauge field gives

$$\sum_{\phi} \kappa(\phi, \omega) \propto p^{|C_{P+1}(Y)|} (1-p)^{|C_{P+1}(X)| - |C_{P+1}(Y)|} |Z^P(Y; \mathbb{F}_N)|. \quad (6.22)$$

Because Y contains the full P -skeleton of X , the group $B^P(Y; \mathbb{F}_N)$ is independent of the choice of open $(P+1)$ -cells, so

$$|Z^P(Y; \mathbb{F}_N)| = |H^P(Y; \mathbb{F}_N)| |B^P(Y; \mathbb{F}_N)| \propto |H^P(Y; \mathbb{F}_N)| = N^{b_P(Y; \mathbb{F}_N)}, \quad (6.23)$$

where b_P is the P th Betty number of Y :

$$b_P(Y; \mathbb{F}_N) := \dim_{\mathbb{F}_N} H_P(Y; \mathbb{F}_N). \quad (6.24)$$

Hence the ω -marginal is, up to normalization,

$$\mu_{X,p,N,P+1}(Y) \propto p^{|C_{P+1}(Y)|} (1-p)^{|C_{P+1}(X)| - |C_{P+1}(Y)|} N^{b_P(Y; \mathbb{F}_N)}. \quad (6.25)$$

This is precisely the $(P+1)$ -dimensional plaquette random-cluster measure on X [36].

Thus the source-free Potts gauge theory and the PRCM are coupled through the joint measure (6.20): the former is the ϕ -marginal and the latter is the Y -marginal. In particular, for prime N , the specialized positive slice of the present spacetime theory inherits the probabilistic structure of the PRCM.

6.2 Middle-dimensional toric specialization

Assume that N is an odd prime and specialize the geometry to the regular self-dual cubical torus

$$X = T_L^{2(P+1)}. \quad (6.26)$$

Then the companion PRCM has plaquette dimension $P+1$, exactly half the ambient dimension.

6.2.1 Sharp homological transition in the PRCM

Let Y denote the random $(P+1)$ -dimensional subcomplex distributed by the PRCM measure (6.25), and let $\iota : Y \hookrightarrow X$ be the inclusion inducing the push-forward on homology $\iota_* : H_{P+1}(Y; \mathbb{F}_N) \rightarrow H_{P+1}(X; \mathbb{F}_N)$. Let A_L and S_L be the events that ι_* is nonzero and surjective respectively.

Theorem 6.1 (Duncan-Schweinhart, [37, Thm. 8]). *If*

$$p_{\text{sd}}(N) := \frac{\sqrt{N}}{1 + \sqrt{N}}, \quad (6.27)$$

then, as $L \rightarrow \infty$:

$$\mu_{X,p,N,P+1}(A_L) \rightarrow 0 \quad \text{for } p < p_{\text{sd}}(N), \quad (6.28)$$

$$\mu_{X,p,N,P+1}(S_L) \rightarrow 1 \quad \text{for } p > p_{\text{sd}}(N), \quad (6.29)$$

Using (6.18), the self-dual random-cluster parameter (6.27) corresponds to the exact critical gauge coupling

$$\beta_{\text{sd}}(N) = -\log(1 - p_{\text{sd}}(N)) = \log(1 + \sqrt{N}). \quad (6.30)$$

6.2.2 Transfer to the gauge theory through the joint measure

The bridge from the PRCM to the source-free Potts gauge theory is furnished by the joint measure (6.20) on the torus X . We write $\nu_{X,\beta_g}^{\text{gauge}}$ for the normalized gauge measure associated with (6.3), and $\langle - \rangle_{X,\beta_g}^{\text{gauge}}$ for expectation with respect to it. Under the joint measure, one may first sample the PRCM subcomplex Y and then sample the gauge field ϕ conditionally on that choice of Y . The conditional law is exactly what transfers homological information about Y into identities for gauge-theoretic Wilson observables.

For a P -cycle $\gamma \in Z_P(X)$, define the associated classical Wilson observable by (the same character as in (3.3))

$$\mathcal{W}_\gamma[\phi] := \chi \left(\int_\gamma \phi \right). \quad (6.31)$$

By the exact coupling via the joint measure κ in (6.20), the conditional law of ϕ given Y is uniform on the cocycle space $Z^P(Y; \mathbb{F}_N)$; see [37, Prop. 21]. Consequently, the finite-volume Wilson/null-homology correspondence of [37, Cor. 29 and Cor. 30], translated into the present notation, gives for every $\gamma, \gamma_1, \gamma_2 \in Z_P(X)$,

$$\mathbb{E}_\kappa[\mathcal{W}_\gamma \mid Y] = \mathbf{1}_{\{[\gamma]=0 \text{ in } H_P(Y; \mathbb{F}_N)\}}, \quad (6.32)$$

$$\mathbb{E}_\kappa[\mathcal{W}_{\gamma_1} \mathcal{W}_{-\gamma_2} \mid Y] = \mathbf{1}_{\{[\gamma_1]=[\gamma_2] \text{ in } H_P(Y; \mathbb{F}_N)\}}. \quad (6.33)$$

Averaging over Y gives

$$\langle \mathcal{W}_\gamma \rangle_{X,\beta_g}^{\text{gauge}} = \mu_{X,p,N,P+1}([\gamma] = 0 \text{ in } H_P(Y; \mathbb{F}_N)), \quad (6.34)$$

$$\langle \mathcal{W}_{\gamma_1} \mathcal{W}_{-\gamma_2} \rangle_{X,\beta_g}^{\text{gauge}} = \mu_{X,p,N,P+1}([\gamma_1] = [\gamma_2] \text{ in } H_P(Y; \mathbb{F}_N)). \quad (6.35)$$

These identities are the precise mechanism by which the PRCM transition is transferred to the gauge theory.

6.2.3 Gauge-theoretic interpretations of the critical coupling

For subsets $S_1, S_2 \subset C_\bullet(X)$, write

$$\text{dist}_X(S_1, S_2) := \min \{ \text{dist}_X(v_1, v_2) : v_i \in C_0(X) \text{ is incident to some cell of } S_i, i = 1, 2 \}, \quad (6.36)$$

where $\text{dist}_X(v_1, v_2)$ is the graph distance in the 1-skeleton of X . Let $\mathcal{P}_L \subset Z_P(X)$ be the finite set of nontrivial toric P -cycles supported on codimension-one subtori of X . Define the gauge event

$$\mathcal{E}_L := \left\{ \phi \in \Omega^P(X) : \exists \gamma_1, \gamma_2 \in \mathcal{P}_L \text{ with } \text{dist}_X(\text{Supp}(\gamma_1), \text{Supp}(\gamma_2)) \geq L/2 - 1 \text{ and } \mathcal{W}_{\gamma_1}[\phi] = \mathcal{W}_{\gamma_2}[\phi] \right\}. \quad (6.37)$$

Proposition 6.2 (Gauge-theoretic consequences of the toric transition). *For $X = T_L^{2(P+1)}$ and N an odd prime:*

1. *If $\Gamma \in Z_P(X)$ represents a nonzero class in $H_P(X; \mathbb{F}_N)$, then*

$$\langle \mathcal{W}_\Gamma \rangle_{X, \beta_g}^{\text{gauge}} = 0 \quad \text{for every } L \text{ and every } \beta_g \geq 0. \quad (6.38)$$

2. *For the event $\mathcal{E}_L$ defined in (6.37),*

$$\nu_{X, \beta_g}^{\text{gauge}}(\mathcal{E}_L) \rightarrow 1 \quad \text{for } \beta_g > \beta_{\text{sd}}(N). \quad (6.39)$$

Proof. Equation (6.38) follows immediately from (6.34). If $[\Gamma] \neq 0$ in $H_P(X; \mathbb{F}_N)$, then $[\Gamma]$ cannot vanish in $H_P(Y; \mathbb{F}_N)$ for any subcomplex $Y \subset X$: if $\Gamma = \partial\rho$ in Y , then the same equality holds in X . Hence the event on the right-hand side of (6.34) is impossible.

For (6.39), (6.29) implies $\mu_{X, p, N, P+1}(A_L) \rightarrow 1$, since $S_L \subset A_L$. On the event A_L , [37, Prop. 36] produces two cycles $\gamma_1, \gamma_2 \in \mathcal{P}_L$ such that

$$\text{dist}_X(\text{Supp}(\gamma_1), \text{Supp}(\gamma_2)) \geq L/2 - 1, \quad [\gamma_1] = [\gamma_2] \text{ in } H_P(Y; \mathbb{F}_N). \quad (6.40)$$

By (6.33),

$$\mathbb{E}_\kappa[\mathcal{W}_{\gamma_1} \mathcal{W}_{-\gamma_2} \mid Y] = 1. \quad (6.41)$$

Since $\mathcal{W}_{\gamma_1} \mathcal{W}_{-\gamma_2}$ takes values in the group of N -th roots of unity, expectation 1 forces

$$\mathcal{W}_{\gamma_1}[\phi] = \mathcal{W}_{\gamma_2}[\phi] \quad \text{for } \kappa\text{-almost every } \phi, \text{ conditioned on } Y. \quad (6.42)$$

Hence $\kappa(\mathcal{E}_L \mid Y) = 1$ on A_L , and therefore

$$\nu_{X, \beta_g}^{\text{gauge}}(\mathcal{E}_L) = \kappa(\mathcal{E}_L) \geq \mu_{X, p, N, P+1}(A_L). \quad (6.43)$$

Using $p = 1 - e^{-\beta_g}$, the convergence (6.39) follows from (6.29) and (6.30). $\square$

Wilson-operator interpretation. Proposition 6.2 shows that the toric transition is not detected by a single nontrivial Wilson one-point function: every fixed nontrivial toric Wilson operator has zero expectation at all couplings. The critical coupling $\beta_{\text{sd}}(N)$ instead marks the onset of macroscopic value-locking. Above $\beta_{\text{sd}}(N)$, the sampled gauge field exhibits with probability tending to one a pair of macroscopically separated nontrivial toric P -cycles whose Wilson evaluations coincide. Below $\beta_{\text{sd}}(N)$, Theorem 6.1 gives (6.28), so the companion PRCM almost surely carries no nontrivial ambient middle-dimensional homology, and the specific PRCM mechanism producing the locking event $\mathcal{E}_L$ is absent with high probability.

Sampling interpretation. The same critical coupling also governs the geometry of the finite-volume plaquette Swendsen-Wang update [41] associated with the joint measure (6.20). Starting from a gauge field ϕ , one first samples Y from the conditional law $\kappa(\cdot \mid \phi)$, and then samples a new gauge field ϕ' from $\kappa(\cdot \mid Y)$. In the present toric middle-dimensional setting, Theorem 6.1 therefore yields a sharp crossover in the sampler. Below $\beta_{\text{sd}}(N)$, the sampled subcomplex Y almost surely has trivial ambient middle-dimensional homology, so the resampling step remains confined to homologically trivial toric data. Above $\beta_{\text{sd}}(N)$, the resampling step accesses nontrivial toric sectors with asymptotically positive probability; in the formulation of [37, §5.1], that probability tends to

$$1 - N^{-\binom{2(P+1)}{P+1}}. \quad (6.44)$$

Because the middle-dimensional toric slice is self-dual, the Kramers-Wannier operator map (5.48) exchanges electric and magnetic toric operators. The same critical coupling may therefore be read, after duality, as the corresponding sharp statement for toric 't Hooft operators and for the appearance of nontrivial toric sectors in the dual sampling dynamics.

7 Conclusion and outlook

We have developed an exact spacetime framework for finite-temperature $\mathbb{Z}_N$ homological codes. The thermal trace, decorated by Wilson and 't Hooft insertions and Euclidean twists, is rewritten at fixed Trotter number as a family of spacetime partition functions labeled by electric and magnetic backgrounds. These partition functions admit an exact reformulation as a gas of closed electric and magnetic defects, or equivalently as a two-species polymer gas of connected closed polymers subject to homological constraints and linking interactions. The same framework also carries an exact higher-form Kramers-Wannier duality exchanging electric and magnetic data, and it contains as a positive specialization the source-free Potts lattice gauge theory and its exact coupling to the PRCM.

The main point is not only that these reformulations exist, but that they provide a workable platform for analysis. In the present paper, we used this platform in two ways. First, by comparing the complex polymer gas to positive same-species hard-core majorants, we obtained a rigorous low-activity regime in which large connected polymers, and in particular homologically nontrivial ones, are exponentially suppressed. Second, on the gauge/PRCM slice, we imported sharp nonperturbative information from the PRCM literature, including the middle-dimensional toric transition and its gauge-theoretic interpretation.

These results leave several natural directions open. One is to sharpen the polymer analysis on the analytic side, for example by exploiting cancellations in the electric-magnetic linking phases or by developing methods adapted directly to the complex two-species gas. A second direction is combinatorial: to improve the counting estimates for rooted support-connected closed polymers entering the low-activity criterion. Since the present constants C_s do not fully exploit closedness, sharper closed-polymer counting bounds would directly sharpen the rigorous low-activity region. Another natural direction is to extend the framework from cell decompositions of closed oriented manifolds to cell decompositions of manifolds with boundary, where one expects boundary conditions and possibly additional boundary degrees of freedom to play an essential role. Another direction is to analyze further special slices and observables within the same spacetime framework, beyond the gauge/PRCM example treated here. A further direction is to extend beyond prime N the genuinely prime-dependent parts of the construction: the generalized linking pairing on arbitrary cycles, the Fourier-resolved sector amplitudes $\widehat{\mathcal{Z}}_M(\ell_m, \ell_e; q_m, q_e)$, and the gauge/PRCM coupling, where coefficient dependence and torsion phenomena should play a more substantial role.

More broadly, the framework developed here suggests that finite-temperature homological codes should be studied not only through Hamiltonians or decoder models, but also through their exact spacetime sector decomposition. In that description, thermal excitations are organized as closed extended defects, topological sectors are encoded by spacetime homology, and duality acts as an exact structural symmetry. From this perspective, questions from lattice gauge theory, homological statistical mechanics, and the combinatorics of closed cycles become directly relevant to quantitative questions about thermal stability. We expect this viewpoint to provide a useful basis for further work at the interface of topological quantum memory, lattice gauge theory, polymer expansions, homological statistical mechanics, and the combinatorics of high-dimensional closed defects.

A Fourier conventions on $\mathbb{Z}_N$

We fix once and for all the primitive N -th root of unity

$$\omega := e^{2\pi i/N}. \quad (\text{A.1})$$

The Pontryagin dual $\widehat{\mathbb{Z}}_N := \text{Hom}(\mathbb{Z}_N, U(1))$ is canonically isomorphic to $\mathbb{Z}_N$ after this choice of ω . For $n = 0, \dots, N-1$, define the elements of $\widehat{\mathbb{Z}}_N$, the characters

$$\chi_n : \mathbb{Z}_N \rightarrow U(1), \quad \chi_n(x) := \omega^{xn}. \quad (\text{A.2})$$

We also define the periodic Kronecker delta

$$\delta_N(x) := \frac{1}{N} \sum_{n=0}^{N-1} \omega^{nx} = \begin{cases} 1, & x \equiv 0 \pmod{N}, \\ 0, & x \not\equiv 0 \pmod{N}. \end{cases} \quad (\text{A.3})$$

From this one immediately obtains the character orthogonality relations

$$\frac{1}{N} \sum_{n \in \widehat{\mathbb{Z}}_N} \chi_n(x) \chi_{-n}(y) = \delta_N(x - y), \quad (\text{A.4})$$

$$\frac{1}{N} \sum_{x \in \mathbb{Z}_N} \chi_m(x) \chi_n(-x) = \delta_N(m - n). \quad (\text{A.5})$$

The characters $\{\chi_n\}_{n \in \widehat{\mathbb{Z}}_N}$ form a basis of the vector space $\text{Fun}(\mathbb{Z}_N, \mathbb{C})$ of complex-valued functions on $\mathbb{Z}_N$. We use the following symmetric normalization for the discrete Fourier transform:

$$f(x) = \frac{1}{\sqrt{N}} \sum_{n \in \widehat{\mathbb{Z}}_N} \hat{f}(n) \chi_n(x), \quad (\text{A.6a})$$

$$\hat{f}(n) = \frac{1}{\sqrt{N}} \sum_{x \in \mathbb{Z}_N} f(x) \chi_n(-x). \quad (\text{A.6b})$$

Orthogonality of the characters implies that these formulas define mutually inverse linear isomorphisms between $\text{Fun}(\mathbb{Z}_N, \mathbb{C})$ and $\text{Fun}(\widehat{\mathbb{Z}}_N, \mathbb{C})$.

B Lattice geometry, duality, and linking

This appendix fixes the geometric and algebraic-topological conventions used throughout the paper. All chain groups, cochain groups, homology groups, and cohomology groups are taken with coefficients in $\mathbb{Z}_N$. In §B.4, in order to define generalized linking numbers for arbitrary cycles, we shall assume N to be prime so that all these groups are $\mathbb{F}_N$ -vector spaces.

We recall that the dimension of the spatial lattice Λ is d , that the code degrees of freedom live on the P -cells of Λ , and that the spacetime complex is

$$\underline{\Lambda} := \Lambda \times S_M^1. \quad (\text{B.1})$$

Here $S_M^1 = \mathbb{Z}/M\mathbb{Z}$ is the cubical decomposition of the circle with vertices $i = 0, \dots, M-1$ and oriented edges $[i, i+1]$.

B.1 Chains, cochains, and the spacetime complex

Let Λ be a finite cell decomposition of a closed oriented d -manifold. For each p , let $C_p(\Lambda)$ denote the set of oriented p -cells of Λ , with the convention

$$C_{-1}(\Lambda) = C_{d+1}(\Lambda) = \emptyset, \quad \Omega^{-1}(\Lambda) = \Omega^{d+1}(\Lambda) = 0. \quad (\text{B.2})$$

For an oriented p -cell b and an oriented $(p-1)$ -cell a , define the incidence number

$$\varepsilon(b, a) := \begin{cases} +1, & a \subset \partial b \text{ and the orientation of } a \text{ agrees with that induced from } \partial b, \\ -1, & a \subset \partial b \text{ and the orientation of } a \text{ is opposite to the induced one,} \\ 0, & a \not\subset \partial b. \end{cases} \quad (\text{B.3})$$

The group of p -chains is free $\mathbb{Z}_N$ -module generated by the p -cells:

$$\mathcal{C}_p(\Lambda) := \mathbb{Z}_N[C_p(\Lambda)], \quad (\text{B.4})$$

and the boundary operator is defined as

$$\partial : \mathcal{C}_{p+1}(\Lambda) \rightarrow \mathcal{C}_p(\Lambda), \quad \partial c := \sum_{b \in C_p(\Lambda)} \varepsilon(c, b) b. \quad (\text{B.5})$$

The incidence numbers satisfy

$$\sum_{b \in C_p(\Lambda)} \varepsilon(c, b) \varepsilon(b, a) = 0 \quad \text{for all } c \in \mathcal{C}_{p+1}(\Lambda), a \in \mathcal{C}_p(\Lambda), \quad (\text{B.6})$$

hence $\partial^2 = 0$.

The space of p -cochains is

$$\Omega^p(\Lambda) := \text{Hom}(\mathcal{C}_p(\Lambda), \mathbb{Z}_N) \cong \mathbb{Z}_N^{C_p(\Lambda)}. \quad (\text{B.7})$$

If $f \in \Omega^p(\Lambda)$ and $\nu = \sum_{b \in C_p(\Lambda)} \nu_b b \in \mathcal{C}_p(\Lambda)$, we write

$$\int_\nu f := f(\nu) = \sum_{b \in C_p(\Lambda)} \nu_b f_b, \quad f_b := f(b) \in \mathbb{Z}_N. \quad (\text{B.8})$$

The coboundary operator

$$d : \Omega^p(\Lambda) \rightarrow \Omega^{p+1}(\Lambda) \quad (\text{B.9})$$

is defined to satisfy the Stokes formula with respect to ∂ :

$$\int_c df = \int_{\partial c} f \quad \text{for every } c \in C_{p+1}(\Lambda). \quad (\text{B.10})$$

In components,

$$(df)_c = \sum_{b \in C_p(\Lambda)} \varepsilon(c, b) f_b. \quad (\text{B.11})$$

Since $\partial^2 = 0$, we also have $d^2 = 0$.

A p -cell of the spacetime lattice $\underline{\Lambda}$ is either horizontal,

$$b(i) := b \times \{i\}, \quad b \in C_p(\Lambda), \quad (\text{B.12})$$

or vertical,

$$\underline{a}(i) := a \times [i, i+1], \quad a \in C_{p-1}(\Lambda). \quad (\text{B.13})$$

Thus the cells of the spacetime lattice decomposes disjointly into horizontal and vertical subsets:

$$C_p(\underline{\Lambda}) = C_p^\parallel(\underline{\Lambda}) \sqcup C_p^\perp(\underline{\Lambda}). \quad (\text{B.14})$$

With the product orientation,

$$\partial(\underline{b}(i)) = (\partial b) \times [i, i+1] + (-1)^p b(i+1) - (-1)^p b(i), \quad (\text{B.15})$$

for every $b \in C_p(\Lambda)$. Accordingly, if

$$\phi \in \Omega^p(\underline{\Lambda}), \quad \phi^\parallel(i) \in \Omega^p(\Lambda), \quad \phi^\perp(i) \in \Omega^{p-1}(\Lambda) \quad (\text{B.16})$$

denote its horizontal and vertical components, then the spacetime coboundary

$$D : \Omega^p(\underline{\Lambda}) \rightarrow \Omega^{p+1}(\underline{\Lambda}) \quad (\text{B.17})$$

is given by

$$(D\phi)_{c(i)} = \left(d\phi^\parallel(i) \right)_c, \quad (\text{B.18a})$$

$$(D\phi)_{\underline{b}(i)} = \left(d\phi^\perp(i) \right)_b + (-1)^p \left(\phi^\parallel(i+1)_b - \phi^\parallel(i)_b \right). \quad (\text{B.18b})$$

A convenient way to write this is

$$(D\phi)^\parallel(i) = d\phi^\parallel(i), \quad (D\phi)^\perp(i) = d\phi^\perp(i) + (-1)^p \Delta_t \phi^\parallel(i), \quad (\text{B.19})$$

where $(\Delta_t \psi)(i) := \psi(i+1) - \psi(i)$. Then

$$(D^2 \phi)^\parallel(i) = d^2 \phi^\parallel(i) = 0, \quad (\text{B.20})$$

and, since $D\phi$ has degree $p+1$,

$$\begin{aligned} (D^2\phi)^\perp(i) &= d((D\phi)^\perp(i)) + (-1)^{p+1}\Delta_t((D\phi)^\parallel(i)) \\ &= d(d\phi^\perp(i) + (-1)^p\Delta_t\phi^\parallel(i)) + (-1)^{p+1}\Delta_t(d\phi^\parallel(i)) \\ &= (-1)^p d\Delta_t\phi^\parallel(i) + (-1)^{p+1}\Delta_t d\phi^\parallel(i) = 0, \end{aligned}$$

because $d^2 = 0$ and d commutes with Δ_t . Hence

$$D^2 = 0. \quad (\text{B.21})$$

For any finite oriented cell complex X , the abelian group of p -cycles Z_p , p -cocycles Z^p , p -boundaries B_p , and p -coboundaries B^p are defined as usual:

$$\begin{aligned} Z_p(X) &:= \ker(\partial : \mathcal{C}_p(X) \rightarrow \mathcal{C}_{p-1}(X)), & Z^p(X) &:= \ker(d : \Omega^p(X) \rightarrow \Omega^{p+1}(X)), \\ B_p(X) &:= \text{im}(\partial : \mathcal{C}_{p+1}(X) \rightarrow \mathcal{C}_p(X)), & B^p(X) &:= \text{im}(d : \Omega^{p-1}(X) \rightarrow \Omega^p(X)). \end{aligned} \quad (\text{B.22})$$

The p th homology and cohomology groups are defined respectively as

$$H_p(X) := Z_p(X)/B_p(X), \quad H^p(X) := Z^p(X)/B^p(X). \quad (\text{B.23})$$

B.2 Chain-cochain duality

The set $C_p(X)$ is a finite basis of $\mathcal{C}_p(X)$, hence there is a module isomorphism

$$\mathcal{P}_X : \Omega^p(X) \xrightarrow{\sim} \mathcal{C}_p(X), \quad \mathcal{P}_X(f) = \sum_{b \in C_p(X)} f_b b. \quad (\text{B.24})$$

Its inverse sends a chain $\sum_{b \in C_p(X)} \nu_b b$ to the cochain with components $\{\nu_b\}_{b \in C_p(X)}$.

Using $\mathcal{P}_X$, define the transpose differential

$$d_X^\top := \mathcal{P}_X^{-1} \partial_X \mathcal{P}_X : \Omega^p(X) \rightarrow \Omega^{p-1}(X). \quad (\text{B.25})$$

If $f \in \Omega^p(X)$, then

$$(d_X^\top f)_a = \sum_{b \in C_p(X)} f_b \varepsilon_X(b, a), \quad a \in C_{p-1}(X). \quad (\text{B.26})$$

We equip $\Omega^p(X)$ with the bilinear pairing

$$\langle f, g \rangle_X := \int_{\mathcal{P}_X(f)} g = \sum_{b \in C_p(X)} f_b g_b, \quad f, g \in \Omega^p(X). \quad (\text{B.27})$$

With respect to this pairing, d_X and $d_X^\top$ are adjoint:

$$\langle d_X^\top f, g \rangle_X = \langle f, d_X g \rangle_X, \quad f \in \Omega^p(X), \quad g \in \Omega^{p-1}(X). \quad (\text{B.28})$$

Now given a finite oriented cell decomposition X of a closed oriented n -manifold, let $X^\vee$ denote its dual cell decomposition. For each $0 \leq r \leq n$, there is a canonical bijection

$$\theta_X : C_r(X^\vee) \xrightarrow{\sim} C_{n-r}(X). \quad (\text{B.29})$$

We choose orientations on the dual cells so that

$$\varepsilon_X(c, b) = \varepsilon_{X^\vee}(\theta_X^{-1}(b), \theta_X^{-1}(c)), \quad (\text{B.30})$$

whenever $b \in C_r(X)$ and $c \in C_{r+1}(X)$. Extending θ_X linearly gives

$$\theta_X : \mathcal{C}_{n-r}(X^\vee) \xrightarrow{\sim} \mathcal{C}_r(X). \quad (\text{B.31})$$

We shall use two induced maps. First, the cochain-level duality map

$$\theta_X^\sharp := \mathcal{P}_X^{-1} \theta_X \mathcal{P}_{X^\vee} : \Omega^{n-r}(X^\vee) \xrightarrow{\sim} \Omega^r(X), \quad (\text{B.32})$$

and second, the map from dual chains to primal cochains

$$\vartheta_X := \mathcal{P}_X^{-1} \circ \theta_X : \mathcal{C}_{n-r}(X^\vee) \rightarrow \Omega^r(X). \quad (\text{B.33})$$

These fit into the commutative diagram

$$\begin{array}{ccc} \Omega^{n-r}(X^\vee) & \xrightarrow{\theta_X^\#} & \Omega^r(X) \\ \mathcal{P}_{X^\vee} \downarrow & \nearrow \vartheta_X & \downarrow \mathcal{P}_X \\ \mathcal{C}_{n-r}(X^\vee) & \xrightarrow{\theta_X} & \mathcal{C}_r(X). \end{array} \quad (\text{B.34})$$

Equivalently,

$$\theta_X^\# = \vartheta_X \circ \mathcal{P}_{X^\vee}, \quad \vartheta_X = \mathcal{P}_X^{-1} \circ \theta_X. \quad (\text{B.35})$$

The key point is that these maps intertwine the relevant differentials.

Proposition B.1 (Intertwining differentials). *Degree-wise, one has*

- (i) $d_X \circ \vartheta_X = \vartheta_X \circ d_{X^\vee}$,
- (ii) $\theta_X^\# \circ d_{X^\vee} = d_X^\top \circ \theta_X^\#$,
- (iii) $d_X \circ \theta_X^\# = \theta_X^\# \circ d_{X^\vee}^\top$.

Proof. 1. Let $c^\vee \in \mathcal{C}_{n-r}(X^\vee)$, so that $\vartheta_X(c^\vee) \in \Omega^r(X)$, and let $a \in C_{r+1}(X)$. Then

$$(d_X \vartheta_X(c^\vee))_a = \sum_{b \in C_r(X)} \varepsilon_X(a, b) \vartheta_X(c^\vee)_b = \varepsilon_X(a, \theta_X(c^\vee)),$$

$$\text{and, } (\vartheta_X(d_{X^\vee} c^\vee))_a = \varepsilon_{X^\vee}(c^\vee, \theta_X^{-1}(a)) = \varepsilon_X(a, \theta_X(c^\vee)),$$

where the last equality is precisely the dual-incidence convention (B.30). This proves Prop. B.1(i).

2. Let $\alpha \in \Omega^{n-r}(X^\vee)$. Then $\theta_X^\# \alpha \in \Omega^r(X)$. For the first identity, evaluate both sides on a cell $a \in C_{r-1}(X)$. We have

$$\left(\theta_X^\# d_{X^\vee} \alpha \right)_a = (d_{X^\vee} \alpha)_{\theta_X^{-1}(a)} = \sum_{c^\vee \in \mathcal{C}_{n-r}(X^\vee)} \varepsilon_{X^\vee}(\theta_X^{-1}(a), c^\vee) \alpha_{c^\vee}.$$

Writing $c = \theta_X(c^\vee) \in C_r(X)$, the dual-incidence convention (B.30) gives $\varepsilon_{X^\vee}(\theta_X^{-1}(a), c^\vee) = \varepsilon_X(c, a)$. Therefore

$$\left(\theta_X^\# d_{X^\vee} \alpha \right)_a = \sum_{c \in C_r(X)} \varepsilon_X(c, a) (\theta_X^\# \alpha)_c = \left(d_X^\top \theta_X^\# \alpha \right)_a. \quad (\text{B.36})$$

This proves Prop. B.1(ii).

3. For the second identity, evaluate both sides on a cell $a \in C_{r+1}(X)$. We get

$$\left(d_X \theta_X^\# \alpha \right)_a = \sum_{b \in C_r(X)} \varepsilon_X(a, b) (\theta_X^\# \alpha)_b = \sum_{b \in C_r(X)} \varepsilon_X(a, b) \alpha_{\theta_X^{-1}(b)}. \quad (\text{B.37})$$

Using the dual-incidence convention again, $\varepsilon_X(a, b) = \varepsilon_{X^\vee}(\theta_X^{-1}(b), \theta_X^{-1}(a))$. Thus

$$\left(d_X \theta_X^\# \alpha \right)_a = \sum_{b \in C_r(X)} \varepsilon_{X^\vee}(\theta_X^{-1}(b), \theta_X^{-1}(a)) \alpha_{\theta_X^{-1}(b)} = (d_{X^\vee}^\top \alpha)_{\theta_X^{-1}(a)} = \left(\theta_X^\# d_{X^\vee}^\top \alpha \right)_a.$$

This proves Prop. B.1(iii). □

Corollary B.2. *ϑ is an isomorphism of complexes.*

Proof. It is a module isomorphism by definition (B.33), since it is a composition of two isomorphisms (B.31) and (B.24). By Prop. B.1(i) it also preserves the differentials. □

B.3 Intersection pairing

For each $0 \leq r \leq n$, define

$$I_X^{(r)} : \mathcal{C}_{n-r}(X^\vee) \times \mathcal{C}_r(X) \rightarrow \mathbb{Z}_N \quad (\text{B.38})$$

by

$$I_X^{(r)}(\Xi^\vee, \Sigma) := \int_{\Sigma} \vartheta_X(\Xi^\vee) = \int_{\Sigma} \mathcal{P}_X^{-1}(\theta_X(\Xi^\vee)). \quad (\text{B.39})$$

On basis elements,

$$I_X^{(r)}(c^\vee, b) = \delta_{\theta_X(c^\vee), b}, \quad c^\vee \in \mathcal{C}_{n-r}(X^\vee), \quad b \in \mathcal{C}_r(X). \quad (\text{B.40})$$

Thus the pairing is nonzero precisely when the cells of the primal and dual lattice correspond to each other under dualization of cell decomposition.

The pairing is compatible with the boundary operators. If

$$\Xi^\vee \in \mathcal{C}_{n-r}(X^\vee), \quad \Sigma \in \mathcal{C}_{r+1}(X), \quad (\text{B.41})$$

then

$$I_X^{(r+1)}(\partial\Xi^\vee, \Sigma) = I_X^{(r)}(\Xi^\vee, \partial\Sigma). \quad (\text{B.42})$$

Indeed,

$$I_X^{(r+1)}(\partial\Xi^\vee, \Sigma) = \int_{\Sigma} \vartheta_X(\partial\Xi^\vee) = \int_{\Sigma} d_X \vartheta_X(\Xi^\vee) = \int_{\partial\Sigma} \vartheta_X(\Xi^\vee) = I_X^{(r)}(\Xi^\vee, \partial\Sigma), \quad (\text{B.43})$$

where we used Prop. B.1(i) and the defining property of d_X (B.10).

B.4 Linking pairing

For two cycles $\nu \in Z_p(X)$ and $\mu^\vee \in Z_{n-p-1}(X^\vee)$, the pair of homology classes

$$([\mu^\vee], [\nu]) \in H_{n-p-1}(X^\vee) \times H_p(X) \quad (\text{B.44})$$

is canonical, but it contains no cycle-level geometric linking information. The purpose of the present appendix is to fix a convenient bilinear pairing on cycles such that the pairing reduces to the standard filling-intersection expression as soon as one of the two cycles bounds.

We assume in this section that N is prime and all chain, cochain, homology, and cohomology groups are finite dimensional vector spaces over

$$\mathbb{F}_N = \mathbb{Z}_N. \quad (\text{B.45})$$

B.4.1 Definition and reduction to ordinary linking

Since all homology groups are finite-dimensional $\mathbb{F}_N$ -vector spaces, the short exact sequences

$$\begin{aligned} 0 \rightarrow B_p(X) \rightarrow Z_p(X) \rightarrow H_p(X) \rightarrow 0, \\ 0 \rightarrow B_{n-p-1}(X^\vee) \rightarrow Z_{n-p-1}(X^\vee) \rightarrow H_{n-p-1}(X^\vee) \rightarrow 0 \end{aligned} \quad (\text{B.46})$$

split. Likewise, because

$$\begin{aligned} 0 \rightarrow Z_{p+1}(X) \rightarrow \mathcal{C}_{p+1}(X) \xrightarrow{\partial} B_p(X) \rightarrow 0, \\ 0 \rightarrow Z_{n-p}(X^\vee) \rightarrow \mathcal{C}_{n-p}(X^\vee) \xrightarrow{\partial} B_{n-p-1}(X^\vee) \rightarrow 0 \end{aligned} \quad (\text{B.47})$$

also split, we may choose once and for all:

1. linear sections

$$s_p : H_p(X) \rightarrow Z_p(X), \quad s_{n-p-1}^\vee : H_{n-p-1}(X^\vee) \rightarrow Z_{n-p-1}(X^\vee), \quad (\text{B.48})$$

2. linear filling operators

$$K_p : B_p(X) \rightarrow \mathcal{C}_{p+1}(X), \quad \partial K_p = \text{id}_{B_p(X)}, \quad (\text{B.49})$$

$$K_{n-p-1}^\vee : B_{n-p-1}(X^\vee) \rightarrow \mathcal{C}_{n-p}(X^\vee), \quad \partial K_{n-p-1}^\vee = \text{id}_{B_{n-p-1}(X^\vee)}, \quad (\text{B.50})$$

3. a bilinear pairing

$$B_H : H_{n-p-1}(X^\vee) \times H_p(X) \rightarrow \mathbb{Z}_N. \quad (\text{B.51})$$

Using the chosen sections, decompose the cycles into homological and exact parts:

$$\begin{aligned} \nu_h &:= s_p([\nu]), & \nu_{ex} &:= \nu - \nu_h \in B_p(X), \\ \mu_h^\vee &:= s_{n-p-1}^\vee([\mu^\vee]), & \mu_{ex}^\vee &:= \mu^\vee - \mu_h^\vee \in B_{n-p-1}(X^\vee). \end{aligned} \quad (\text{B.52})$$

Thus

$$\nu = \nu_h + \nu_{ex}, \quad \mu^\vee = \mu_h^\vee + \mu_{ex}^\vee. \quad (\text{B.53})$$

We then define the generalized linking pairing

$$\text{Lk}_X : Z_{n-p-1}(X^\vee) \times Z_p(X) \rightarrow \mathbb{Z}_N \quad (\text{B.54})$$

by

$$\text{Lk}_X(\mu^\vee, \nu) := I_X^{(p+1)}(\mu_h^\vee, K_p(\nu_{ex})) + I_X^{(p)}(K_{n-p-1}^\vee(\mu_{ex}^\vee), \nu_h) + B_H([\mu^\vee], [\nu]). \quad (\text{B.55})$$

By construction, Lk_X is bilinear.

The three terms in (B.55) have distinct roles. The first fills the exact part of the primal cycle and intersects that filling with the full dual cycle. The second fills the exact part of the dual cycle and intersects that filling with the chosen homology representative of the primal cycle. The third depends only on the homology classes and prescribes the residual purely homological coupling. One may rewrite (B.55) in the equivalent form

$$\text{Lk}_X(\mu^\vee, \nu) = I_X^{(p+1)}(\mu_h^\vee, K_p(\nu_{ex})) + I_X^{(p)}(K_{n-p-1}^\vee(\mu_{ex}^\vee), \nu) + B_H([\mu^\vee], [\nu]). \quad (\text{B.56})$$

Indeed,

$$\begin{aligned} I_X^{(p+1)}(\mu_{ex}^\vee, K_p(\nu_{ex})) &= I_X^{(p+1)}(\partial K_{n-p-1}^\vee(\mu_{ex}^\vee), K_p(\nu_{ex})) && \text{using (B.50)} \\ &= I_X^{(p)}(K_{n-p-1}^\vee(\mu_{ex}^\vee), \partial K_p(\nu_{ex})) && \text{using (B.42)} \\ &= I_X^{(p)}(K_{n-p-1}^\vee(\mu_{ex}^\vee), \nu_{ex}). && \text{using (B.49)} \end{aligned} \quad (\text{B.57})$$

(B.56) then follows from (B.55) by expanding $\mu^\vee$ and ν as in (B.53).

The importance of (B.55) is that it reduces to ordinary linking as soon as one of the two cycles bounds. If $\nu \in B_p(X)$, then $[\nu] = 0$, hence $\nu_h = 0$ and $\nu_{ex} = \nu$, so

$$\text{Lk}_X(\mu^\vee, \nu) = I_X^{(p+1)}(\mu_h^\vee, K_p(\nu)). \quad (\text{B.58})$$

Thus one simply fills ν and intersects that filling with $\mu^\vee$. Similarly, if $\mu^\vee \in B_{n-p-1}(X^\vee)$, then $[\mu^\vee] = 0$, hence $\mu_h^\vee = 0$ and $\mu_{ex}^\vee = \mu^\vee$, so

$$\text{Lk}_X(\mu^\vee, \nu) = I_X^{(p)}(K_{n-p-1}^\vee(\mu^\vee), \nu). \quad (\text{B.59})$$

Thus one may instead fill $\mu^\vee$ and intersect that filling with ν . In particular, if both $\nu \in B_p(X)$ and $\mu^\vee \in B_{n-p-1}(X^\vee)$, then

$$\text{Lk}_X(\mu^\vee, \nu) = I_X^{(p+1)}(\mu^\vee, K_p(\nu)) = I_X^{(p)}(K_{n-p-1}^\vee(\mu^\vee), \nu), \quad (\text{B.60})$$

which is exactly the ordinary linking/intersection number obtained by filling either cycle and intersecting with the other.

It is also useful to isolate the genuinely boundary-linking part:

$$\text{Lk}_X^{\text{ex}}(\mu^\vee, \nu) := I_X^{(p+1)}(\mu_{ex}^\vee, K_p(\nu_{ex})) = I_X^{(p)}(K_{n-p-1}^\vee(\mu_{ex}^\vee), \nu_{ex}). \quad (\text{B.61})$$

Then

$$\text{Lk}_X(\mu^\vee, \nu) = \text{Lk}_X^{\text{ex}}(\mu^\vee, \nu) + I_X^{(p+1)}(\mu_h^\vee, K_p(\nu_{ex})) + I_X^{(p)}(K_{n-p-1}^\vee(\mu_{ex}^\vee), \nu_h) + B_H([\mu^\vee], [\nu]). \quad (\text{B.62})$$

The first term depends only on the exact parts, the last only on the homology classes, and the middle two terms measure the coupling between exact data and the chosen homology representatives.

B.4.2 Dependence on the auxiliary choices

The pairing Lk_X is not canonical. It depends on the chosen sections, filling operators, and bilinear homological correction. For the applications in the main text, the point is that once the filling operators K_p and $K_{n-p-1}^\vee$ are fixed, changing the other conventions changes Lk_X only by a bilinear function of the homology classes.

Proposition B.3. *Fix a single choice of the filling operators (B.49) and (B.50). Let*

$$(s_p, s_{n-p-1}^\vee, B_H) \quad \text{and} \quad (\tilde{s}_p, \tilde{s}_{n-p-1}^\vee, \tilde{B}_H) \quad (\text{B.63})$$

be two choices of sections (B.48) and homological correction term (B.51), and let Lk_X and $\tilde{\text{Lk}}_X$ be the corresponding generalized linking pairings (B.55). Then:

(i) *There exists a bilinear map*

$$C_H : H_{n-p-1}(X^\vee) \times H_p(X) \rightarrow \mathbb{Z}_N \quad (\text{B.64})$$

such that

$$\tilde{\text{Lk}}_X(\mu^\vee, \nu) - \text{Lk}_X(\mu^\vee, \nu) = C_H([\mu^\vee], [\nu]) \quad (\text{B.65})$$

for all $\mu^\vee \in Z_{n-p-1}(X^\vee)$ and $\nu \in Z_p(X)$. In particular, the two pairings agree whenever either $\mu^\vee$ or ν is a boundary.

(ii) *If $\mu^\vee \in B_{n-p-1}(X^\vee)$ and $\nu \in B_p(X)$, then*

$$\text{Lk}_X(\mu^\vee, \nu) = I_X^{(p+1)}(\mu^\vee, \Sigma) = I_X^{(p)}(T^\vee, \nu), \quad \partial\Sigma = \nu, \quad \partial T^\vee = \mu^\vee. \quad (\text{B.66})$$

This value is independent of all auxiliary choices used in the definition of Lk_X , including the sections, the homological correction term, and the filling chains $\Sigma, T^\vee$. In particular, the linking number of two boundaries is canonical and does not require the assumption that N be prime.

Proof. Set

$$\begin{aligned} \alpha &:= \tilde{s}_p - s_p : H_p(X) \rightarrow Z_p(X), \\ \beta &:= \tilde{s}_{n-p-1}^\vee - s_{n-p-1}^\vee : H_{n-p-1}(X^\vee) \rightarrow Z_{n-p-1}(X^\vee), \end{aligned} \quad (\text{B.67})$$

and

$$C_H^0 := \tilde{B}_H - B_H : H_{n-p-1}(X^\vee) \times H_p(X) \rightarrow \mathbb{Z}_N. \quad (\text{B.68})$$

Since both s_p and $\tilde{s}_p$ are sections of $Z_p(X) \twoheadrightarrow H_p(X)$, the image of α lies in $B_p(X)$. Likewise, the image of β lies in $B_{n-p-1}(X^\vee)$.

Writing the homological and exact parts defined by the two choices of sections, we have

$$\begin{aligned} \tilde{\nu}_h &= \nu_h + \alpha([\nu]), & \tilde{\nu}_{ex} &= \nu_{ex} - \alpha([\nu]), \\ \tilde{\mu}_h^\vee &= \mu_h^\vee + \beta([\mu^\vee]), & \tilde{\mu}_{ex}^\vee &= \mu_{ex}^\vee - \beta([\mu^\vee]). \end{aligned} \quad (\text{B.69})$$

Substituting these into the definition of $\tilde{\text{Lk}}_X$, expanding, and using (B.42) to move fillings of exact terms from one side to the other, we obtain

$$\begin{aligned} \tilde{\text{Lk}}_X(\mu^\vee, \nu) - \text{Lk}_X(\mu^\vee, \nu) &= -I_X^{(p+1)}(\mu_h^\vee, K_p(\alpha([\nu]))) - I_X^{(p)}(K_{n-p-1}^\vee(\beta([\mu^\vee])), \nu_h) \\ &\quad - I_X^{(p)}(K_{n-p-1}^\vee(\beta([\mu^\vee])), \alpha([\nu])) + C_H^0([\mu^\vee], [\nu]). \end{aligned} \quad (\text{B.70})$$

Now $\mu_h^\vee = s_{n-p-1}^\vee([\mu^\vee])$ and $\nu_h = s_p([\nu])$, so the right-hand side depends only on $[\mu^\vee]$ and $[\nu]$, and it is bilinear because all maps appearing in (B.70) are linear or bilinear. This proves (B.65) and therefore (i). If either $[\mu^\vee] = 0$ or $[\nu] = 0$, then all terms on the right-hand side of (B.70) vanish, so the two pairings agree whenever either argument is a boundary.

For (ii), if $\mu^\vee \in B_{n-p-1}(X^\vee)$ and $\nu \in B_p(X)$, then (B.60) gives

$$\text{Lk}_X(\mu^\vee, \nu) = I_X^{(p+1)}(\mu^\vee, \Sigma) = I_X^{(p)}(T^\vee, \nu), \quad \partial\Sigma = \nu, \quad \partial T^\vee = \mu^\vee. \quad (\text{B.71})$$

By (i), this value is independent of the chosen sections and homological correction term, because both homology classes vanish. It remains only to check independence of the fillings. If $\Sigma, \Sigma' \in \mathcal{C}_{p+1}(X)$ both satisfy $\partial\Sigma = \partial\Sigma' = \nu$, then $\Sigma - \Sigma' \in Z_{p+1}(X)$, so using $\mu^\vee = \partial T^\vee$ and (B.42),

$$I_X^{(p+1)}(\mu^\vee, \Sigma - \Sigma') = I_X^{(p+1)}(\partial T^\vee, \Sigma - \Sigma') = I_X^{(p)}(T^\vee, \partial(\Sigma - \Sigma')) = 0. \quad (\text{B.72})$$

Hence $I_X^{(p+1)}(\mu^\vee, \Sigma)$ is independent of Σ . The argument for $I_X^{(p)}(T^\vee, \nu)$ is identical. Therefore the common value (B.66) is canonical. Since its definition uses only ordinary filling-intersection of two boundaries, it does not depend on the prime- N hypothesis introduced to construct a generalized pairing on arbitrary cycles. $\square$

Proposition B.3 identifies the precise source of the ambiguity in the generalized linking pairing. Changing the cycle-level convention does not alter the ordinary linking content of boundary interactions; it changes only a purely homological bilinear correction.

C Supporting computations

This appendix gives the finite- M derivation of the spacetime representation used in §2. The calculation is completely finite-dimensional. Its purpose is to justify the local kernels, the spacetime P -cochain packaging, and the effect of the spatial and Euclidean twist insertions.

Throughout this appendix we use the $\hat{Z}$ -eigenbasis of the Hilbert space

$$\mathcal{H} = \bigotimes_{b \in C_P(\Lambda)} \mathcal{H}_b, \quad |s\rangle = \bigotimes_{b \in C_P(\Lambda)} |s_b\rangle, \quad s \in \Omega^P(\Lambda) \cong \mathbb{Z}_N^{C_P(\Lambda)}. \quad (\text{C.1})$$

In this basis,

$$\hat{Z}_b |s\rangle = \omega^{s_b} |s\rangle, \quad \hat{X}_b |s\rangle = |s + \delta_b\rangle, \quad \omega = e^{2\pi i/N}, \quad (\text{C.2})$$

where $\delta_b \in \Omega^P(\Lambda)$ is the cochain whose value on b is 1 and whose value on every other P -cell is 0.

C.1 Proof of Prop. 2.1

1. It is enough to show that $\hat{\mathcal{W}}_\nu$ and $\hat{\mathcal{T}}_{\mu^\vee}$ commute with all stabilizer generators $\hat{A}_a$ and $\hat{B}_c$.

First, $\hat{\mathcal{W}}_\nu$ is a product of $\hat{Z}$ -operators, so it commutes with every $\hat{B}_c$. For $a \in C_{P-1}(\Lambda)$, using the Weyl relations (2.5) we get

$$\hat{A}_a \hat{\mathcal{W}}_\nu = \omega^{-(-1)^P \sum_{b \in C_P(\Lambda)} \varepsilon(b,a) \nu_b} \hat{\mathcal{W}}_\nu \hat{A}_a. \quad (\text{C.3})$$

The exponent is the coefficient of a in $\partial\nu$ (see (B.5)). Since $\nu \in Z_P(\Lambda)$, we have $\partial\nu = 0$, and therefore $[\hat{A}_a, \hat{\mathcal{W}}_\nu] = 0$.

Similarly, $\hat{\mathcal{T}}_{\mu^\vee}$ is a product of $\hat{X}$ -operators, so it commutes with every $\hat{A}_a$. Set $\eta := \vartheta_\Lambda(\mu^\vee) \in \Omega^P(\Lambda)$. For $c \in C_{P+1}(\Lambda)$, using the Weyl relations again, we find

$$\hat{B}_c \hat{\mathcal{T}}_{\mu^\vee} = \omega^{(-1)^P \sum_{b \in C_P(\Lambda)} \varepsilon(c,b) \eta_b} \hat{\mathcal{T}}_{\mu^\vee} \hat{B}_c = \omega^{(-1)^P (d\eta)_c} \hat{\mathcal{T}}_{\mu^\vee} \hat{B}_c. \quad (\text{C.4})$$

By the primal-dual intertwining relation (Prop. B.1(i)),

$$d\eta = d\vartheta_\Lambda(\mu^\vee) = \vartheta_\Lambda(\partial\mu^\vee) = 0, \quad (\text{C.5})$$

because $\mu^\vee \in Z_{d-P}(\Lambda^\vee)$. Thus $[\hat{B}_c, \hat{\mathcal{T}}_{\mu^\vee}] = 0$.

Therefore both $\hat{\mathcal{W}}_\nu$ and $\hat{\mathcal{T}}_{\mu^\vee}$ preserve the common +1-eigenspace of the stabilizers, namely $\mathcal{H}_{\text{code}}$, proving Prop. (i).

2. If $\nu = \partial C$, with $C = \sum_c C_c c$, then $\nu_b = \sum_{c \in C_{P+1}(\Lambda)} C_c \varepsilon(c, b)$. Therefore

$$\hat{\mathcal{W}}_\nu = \prod_{b \in C_P(\Lambda)} \hat{Z}_b^{\nu_b} = \prod_{c \in C_{P+1}(\Lambda)} \hat{B}_c^{C_c}. \quad (\text{C.6})$$

This proves Prop. 2.1(ii).

3. Similarly, if $\mu^\vee = \partial \Xi^\vee$ for some $(d - P + 1)$ -chain $\Xi^\vee$, set $\eta := \vartheta_\Lambda(\Xi^\vee) \in \Omega^{P-1}(\Lambda)$. Then by the primal-dual intertwining relation, $\vartheta_\Lambda(\mu^\vee) = d\eta$. Hence

$$\hat{\mathcal{T}}_{\mu^\vee} = \prod_{b \in C_P(\Lambda)} \hat{X}_b^{(-1)^P (d\eta)_b} = \prod_{a \in C_{P-1}(\Lambda)} \hat{A}_a^{\eta_a}, \quad (\text{C.7})$$

which proves Prop. 2.1(iii).

4. Finally, using the Weyl relation (2.5) we get

$$\hat{X}_b^{(-1)^P I_\Lambda(\mu^\vee, b)} \hat{Z}_b^{\nu_b} = \omega^{(-1)^P I_\Lambda(\mu^\vee, b) \nu_b} \hat{Z}_b^{\nu_b} \hat{X}_b^{(-1)^P I_\Lambda(\mu^\vee, b)}. \quad (\text{C.8})$$

Multiplying over $b \in C_P(\Lambda)$ gives Prop. 2.1(iv)

$$\hat{\mathcal{T}}_{\mu^\vee} \hat{\mathcal{W}}_\nu = \omega^{(-1)^P \sum_b I_\Lambda(\mu^\vee, b) \nu_b} \hat{\mathcal{W}}_\nu \hat{\mathcal{T}}_{\mu^\vee} = \omega^{(-1)^P I_\Lambda(\mu^\vee, \nu)} \hat{\mathcal{W}}_\nu \hat{\mathcal{T}}_{\mu^\vee}. \quad (\text{C.9})$$

C.2 Untwisted Trotter kernels

The fixed- M Trotterized trace is (2.20)

$$Z_M = \text{tr}_{\mathcal{H}} \left[\left(e^{-\beta \hat{H}_z / M} e^{-\beta \hat{H}_x / M} \right)^M \right]. \quad (\text{C.10})$$

Inserting M resolutions of the identity in the $\hat{Z}$ -basis and using the periodic convention $s(M) = s(0)$, we obtain

$$Z_M = \sum_{s(0), \dots, s(M-1) \in \Omega^P(\Lambda)} \prod_{i=0}^{M-1} \left\langle s(i) \left| e^{-\beta \hat{H}_z / M} \right| s(i) \right\rangle \left\langle s(i) \left| e^{-\beta \hat{H}_x / M} \right| s(i+1) \right\rangle. \quad (\text{C.11})$$

We now compute the two one-slice kernels.

C.2.1 The diagonal Z -kernel

For $c \in C_{P+1}(\Lambda)$, the Z -stabilizer is $\hat{B}_c = \prod_{b \in C_c} \hat{Z}_b^{\varepsilon(c, b)}$ (see (2.7)). Therefore

$$\hat{B}_c |s\rangle = \omega^{(ds)_c} |s\rangle, \quad (ds)_c = \sum_{b \in C_P(\Lambda)} \varepsilon(c, b) s_b. \quad (\text{C.12})$$

It follows that the Z -projector from (2.8) acts on the Z -eigenbasis (C.2) as

$$\hat{\mathcal{B}}_c |s\rangle = \frac{1}{N} \sum_{m \in \mathbb{Z}_N} \omega^{m(ds)_c} |s\rangle = \delta_N((ds)_c) |s\rangle. \quad (\text{C.13})$$

The Z -source term (see (2.10)) is also diagonal:

$$\exp \left(\frac{\beta}{M} \sum_{n \in \mathbb{Z}_N} h_b^{(n)} \hat{Z}_b^n \right) |s\rangle = \exp \left(\frac{\beta}{M} \sum_{n \in \mathbb{Z}_N} h_b^{(n)} \omega^{ns_b} \right) |s\rangle. \quad (\text{C.14})$$

Using the definition of $\hat{H}_z$ (2.14), we obtain

$$\left\langle s(i) \left| e^{-\beta \hat{H}_z / M} \right| s(i) \right\rangle = \prod_{c \in C_{P+1}(\Lambda)} W_{c(i)}^\parallel ((ds(i))_c) \prod_{b \in C_P(\Lambda)} V_{b(i)}^\parallel (s(i)_b), \quad (\text{C.15})$$

where the weight functions are defined in (2.21) as

$$W_{c(i)}^\parallel(x) = \exp \left(\frac{\beta K_c}{M} \delta_N(x) \right) = 1 + \left(e^{\beta K_c / M} - 1 \right) \delta_N(x), \quad (\text{C.16a})$$

$$V_{b(i)}^\parallel(x) = \exp \left(\frac{\beta}{M} \sum_{n \in \mathbb{Z}_N} h_b^{(n)} \omega^{nx} \right). \quad (\text{C.16b})$$

C.2.2 The off-diagonal X -kernel

We next compute the kernel of $e^{-\beta \hat{H}_x/M}$. Since all terms in $\hat{H}_x$ commute, we may separate the $\hat{A}$ -projector part from the local $\hat{X}$ -source part.

For each $a \in C_{P-1}(\Lambda)$, the operator $\hat{A}_a$ (2.8) is a projector, so

$$e^{(\beta J_a/M) \hat{A}_a} = 1 + \left(e^{\beta J_a/M} - 1 \right) \hat{A}_a = \sum_{n_a \in \mathbb{Z}_N} \left[\delta_N(n_a) + \frac{e^{\beta J_a/M} - 1}{N} \right] \hat{A}_a^{n_a}. \quad (\text{C.17})$$

Using $\hat{A}_a = \prod_{b \supset a} \hat{X}_b^{(-1)^P \varepsilon(b,a)}$, we get, for $n \in \Omega^{P-1}(\Lambda)$,

$$\prod_{a \in C_{P-1}(\Lambda)} \hat{A}_a^{n_a} = \prod_{b \in C_P(\Lambda)} \hat{X}_b^{(-1)^P (dn)_b}, \quad \prod_{a \in C_{P-1}(\Lambda)} \hat{A}_a^{n_a} |s\rangle = |s + (-1)^P dn\rangle. \quad (\text{C.18})$$

Therefore

$$\begin{aligned} \left\langle t \left| e^{\frac{\beta}{M} \sum_{a \in C_{P-1}(\Lambda)} J_a \hat{A}_a} \right| s \right\rangle &= \sum_{n \in \Omega^{P-1}(\Lambda)} \prod_{a \in C_{P-1}(\Lambda)} \left[\delta_N(n_a) + \frac{e^{\beta J_a/M} - 1}{N} \right] \\ &\quad \times \prod_{b \in C_P(\Lambda)} \delta_N(t_b - s_b - (-1)^P (dn)_b). \end{aligned} \quad (\text{C.19})$$

For the local X -source, introduce the spectral projectors of $\hat{X}$:

$$\hat{\Pi}_j^{(X)} := \frac{1}{N} \sum_{n=0}^{N-1} \omega^{-jn} \hat{X}^n, \quad j \in \mathbb{Z}_N. \quad (\text{C.20})$$

They satisfy

$$\hat{X} \hat{\Pi}_j^{(X)} = \omega^j \hat{\Pi}_j^{(X)}, \quad \sum_{j \in \mathbb{Z}_N} \hat{\Pi}_j^{(X)} = \hat{1}, \quad \left\langle s_b \left| \hat{\Pi}_j^{(X)} \right| t_b \right\rangle = \frac{1}{N} \omega^{j(t_b - s_b)}. \quad (\text{C.21})$$

Consequently,

$$\left\langle s \left| e^{\frac{\beta}{M} \sum_{b \in C_P(\Lambda)} \sum_{n \in \mathbb{Z}_N} g_b^{(n)} \hat{X}_b^n} \right| t \right\rangle = \prod_{b \in C_P(\Lambda)} \sum_{j \in \mathbb{Z}_N} \exp \left(\frac{\beta}{M} \sum_{n \in \mathbb{Z}_N} g_b^{(n)} \omega^{nj} \right) \frac{\omega^{j(t_b - s_b)}}{N}. \quad (\text{C.22})$$

Convolving (C.19) and (C.22), and using the delta-functions to eliminate the intermediate field t , gives

$$\begin{aligned} \left\langle s(i) \left| e^{-\beta \hat{H}_x/M} \right| s(i+1) \right\rangle &= \sum_{n(i) \in \Omega^{P-1}(\Lambda)} \prod_{b \in C_P(\Lambda)} W_{\underline{b}(i)}^\perp \left((dn(i))_b + (-1)^P (s(i+1)_b - s(i)_b) \right) \\ &\quad \times \prod_{a \in C_{P-1}(\Lambda)} V_{\underline{a}(i)}^\perp (n(i)_a), \end{aligned} \quad (\text{C.23})$$

where the weight functions are defined in (2.22) as

$$V_{\underline{a}(i)}^\perp(x) = \delta_N(x) + \frac{e^{\beta J_a/M} - 1}{N}, \quad (\text{C.24a})$$

$$W_{\underline{b}(i)}^\perp(x) = \sum_{j \in \mathbb{Z}_N} \exp \left(\frac{\beta}{M} \sum_{n \in \mathbb{Z}_N} g_b^{(n)} \omega^{nj} \right) \frac{\omega^{(-1)^P jx}}{N}. \quad (\text{C.24b})$$

C.2.3 Packaging into a spacetime P -cochain

Define a spacetime P -cochain $\phi \in \Omega^P(\underline{\Lambda})$ by

$$\phi_{b(i)} := s(i)_b, \quad \phi_{\underline{a}(i)} := n(i)_a, \quad (\text{C.25})$$

where $b(i) = b \times \{i\}$ is a horizontal P -cell and $\underline{a}(i) = a \times [i, i+1]$ is a vertical P -cell, as defined in (B.12) and (B.13) respectively. Substituting (C.15) and (C.23) into (C.11) and identifying the spacetime differential (B.18) in the arguments of W we get the untwisted spacetime partition function

$$Z_M = \sum_{\phi \in \Omega^P(\underline{\Lambda})} \prod_{c \in C_{P+1}(\underline{\Lambda})} W_c((D\phi)_c) \prod_{u \in C_P(\underline{\Lambda})} V_u(\phi_u). \quad (\text{C.26})$$

Here W_c denotes $W^\parallel$ on horizontal $(P+1)$ -cells and $W^\perp$ on vertical $(P+1)$ -cells; similarly V_u denotes $V^\parallel$ on horizontal P -cells and $V^\perp$ on vertical P -cells.

C.3 Spatial Wilson and 't Hooft insertions

We next insert the spatial Wilson operator and 't Hooft operators (2.17). The decorated trace with only these spatial insertions is

$$Z_M^{\text{sp}}(\mu^\vee, \nu) := \text{tr}_{\mathcal{H}} \left[e^{-\beta \hat{H}_z/M} \hat{\mathcal{W}}_\nu e^{-\beta \hat{H}_x/M} \hat{\mathcal{T}}_{\mu^\vee} \left(e^{-\beta \hat{H}_z/M} e^{-\beta \hat{H}_x/M} \right)^{M-1} \right]. \quad (\text{C.27})$$

The Wilson operator is diagonal:

$$\langle s(0) | \hat{\mathcal{W}}_\nu | s(0) \rangle = \prod_{b \in C_P(\Lambda)} \omega^{\nu_b s(0)_b} = \chi \left(\int_\nu s(0) \right). \quad (\text{C.28})$$

Equivalently, after packaging into ϕ , this is $\chi \left(\int_{\nu(0)} \phi \right)$, where $\nu(0) \in Z_P(\underline{\Lambda})$ is the horizontal lift of ν to the initial time slice.

Using the definition of the 't Hooft operator $\hat{\mathcal{T}}_{\mu^\vee} = \prod_{b \in C_P(\Lambda)} \hat{X}_b^{(-1)^P I_\Lambda(\mu^\vee, b)}$, and noting that $I_\Lambda(\mu^\vee, b) = \mu_{\theta_\Lambda^{-1}(b)}^\vee = \vartheta_\Lambda(\mu^\vee)_b$ (see (B.40), (B.33), and (B.24)) we find

$$\hat{\mathcal{T}}_{\mu^\vee} |s\rangle = |s + (-1)^P \vartheta_\Lambda(\mu^\vee)\rangle. \quad (\text{C.29})$$

Since the $\hat{X}$ -kernel depends on $s(i)$ and $s(i+1)$ through the combination

$$(dn(i))_b + (-1)^P (s(i+1)_b - s(i)_b), \quad (\text{C.30})$$

the insertion of $\hat{\mathcal{T}}_{\mu^\vee}$ on the first slab changes the vertical argument by

$$(dn(i))_b + (-1)^P (s(i+1)_b - s(i)_b) \mapsto (dn(i))_b + (-1)^P (s(i+1)_b - s(i)_b) + \delta_{i,0}(\vartheta_\Lambda(\mu^\vee))_b. \quad (\text{C.31})$$

Let $\mu^\vee(0) \in Z_{d-P}(\underline{\Lambda}^\vee)$ denote the dual spacetime cycle supported on the first dual slab and characterized by

$$(\vartheta_\Lambda(\mu^\vee(0)))_{\underline{b}(0)} = (\vartheta_\Lambda(\mu^\vee))_b, \quad (\vartheta_\Lambda(\mu^\vee(0)))_{c(i)} = 0, \quad (\vartheta_\Lambda(\mu^\vee(0)))_{\underline{b}(i)} = 0 \quad (i \neq 0). \quad (\text{C.32})$$

Then (C.31) is exactly the replacement

$$D\phi \mapsto D\phi + \vartheta_\Lambda(\mu^\vee(0)). \quad (\text{C.33})$$

Hence

$$Z_M^{\text{sp}}(\mu^\vee, \nu) = \sum_{\phi \in \Omega^P(\underline{\Lambda})} \chi \left(\int_{\nu(0)} \phi \right) \prod_{c \in C_{P+1}(\underline{\Lambda})} W_c((D\phi + \vartheta_\Lambda(\mu^\vee(0)))_c) \prod_{u \in C_P(\underline{\Lambda})} V_u(\phi_u). \quad (\text{C.34})$$

C.4 Euclidean twists

C.4.1 Electric twist

For $\alpha = \sum_{a \in C_{P-1}(\Lambda)} \alpha_a a \in Z_{P-1}(\Lambda)$, using the orthogonality relations (2.30) of the spectral projectors $\hat{\Pi}_{a,j}^{(A)} = \frac{1}{N} \sum_{m=0}^{N-1} \omega^{-jm} \hat{A}_a^m$ and the definition of the Euclidean twist operator (2.32) we get

$$\begin{aligned} \hat{U}_{e,a}(\alpha_a) e^{(\beta J_a/M) \hat{A}_a} &= \sum_{j \in \mathbb{Z}_N} \lambda_{a,j+\alpha_a} \hat{\Pi}_{a,j} \\ &= \sum_{n_a \in \mathbb{Z}_N} \left[\delta_N(n_a) + \frac{e^{\beta J_a/M} - 1}{N} \right] \omega^{\alpha_a n_a} \hat{A}_a^{n_a}. \end{aligned} \quad (\text{C.35})$$

The factor $\omega^{\alpha_a n_a}$ is harmless on the $\delta_N(n_a)$ term, since $\delta_N(n_a)\omega^{\alpha_a n_a} = \delta_N(n_a)$. Therefore, relative to the untwisted X -kernel, inserting $\hat{U}_e(\alpha)$ on the i -th slab contributes the factor

$$\prod_{a \in C_{P-1}(\Lambda)} \omega^{\alpha_a n(i)_a} = \chi \left(\sum_{a \in C_{P-1}(\Lambda)} \alpha_a n(i)_a \right). \quad (\text{C.36})$$

Under the spacetime packaging (C.25), this becomes $\chi \left(\int_{\underline{\alpha}(i)} \phi \right)$. Multiplying over all M slabs gives

$$\prod_{i=0}^{M-1} \chi \left(\int_{\underline{\alpha}(i)} \phi \right) = \chi \left(\int_{\text{Sus}(\alpha)} \phi \right), \quad (\text{C.37})$$

where $\text{Sus}(\alpha) = \sum_{i=0}^{M-1} \underline{\alpha}(i) \in Z_P(\underline{\Lambda})$ as in (2.34).

C.4.2 Magnetic twist

For $c \in C_{P+1}(\Lambda)$ the spectral projectors $\hat{\Pi}_{c,j}^{(B)} = \frac{1}{N} \sum_{m=0}^{N-1} \omega^{-jm} \hat{B}_c^m$ act on the Z -eigenbasis (C.2) as

$$\hat{\Pi}_{c,j}^{(B)} |s\rangle = \delta_N((ds)_c - j) |s\rangle. \quad (\text{C.38})$$

Therefore, for $\beta^\vee \in Z_{d-P-1}(\Lambda^\vee)$ with $\beta_c = (\vartheta_\Lambda(\beta^\vee))_c$, the magnetic twist operator (2.40) acts on the Z -eigenbasis as

$$\hat{U}_{m,c}(\beta_c) e^{(\beta K_c/M) \hat{B}_c} |s\rangle = \rho_{c,(ds)_c + \beta_c} |s\rangle = \exp \left(\frac{\beta K_c}{M} \delta_N((ds)_c + \beta_c) \right) |s\rangle. \quad (\text{C.39})$$

Hence

$$\left\langle s(i) \left| \hat{U}_m(\beta^\vee) e^{-\beta \hat{H}_z/M} \right| s(i) \right\rangle = \prod_{c \in C_{P+1}(\Lambda)} W_{c(i)}^\parallel ((ds(i))_c + (\vartheta_\Lambda(\beta^\vee))_c) \prod_{b \in C_P(\Lambda)} V_{b(i)}^\parallel (s(i)_b). \quad (\text{C.40})$$

Thus the magnetic twist is precisely the horizontal cocycle shift

$$D\phi \mapsto D\phi + \vartheta_{\underline{\Lambda}}(\text{Sus}(\beta^\vee)), \quad (\text{C.41})$$

where the suspended dual cycle is defined as in (2.42).

C.5 Fully decorated trace

Consider the fully decorated trace (2.44). All X -type operators appearing in $\hat{T}_{\mu^\vee} \hat{U}_e(\alpha) e^{-\beta \hat{H}_x/M}$ commute with one another, and all Z -type operators appearing in $\hat{\mathcal{W}}_\nu \hat{U}_m(\beta^\vee) e^{-\beta \hat{H}_z/M}$ commute with one another. Thus the preceding kernel calculations apply directly.

The decorated X -kernel is

$$\begin{aligned} & \left\langle s(i) \left| \hat{T}_{\mu^\vee}^{\delta_{i,0}} \hat{U}_e(\alpha) e^{-\beta \hat{H}_x/M} \right| s(i+1) \right\rangle \\ &= \sum_{n(i) \in \Omega^{P-1}(\Lambda)} \chi \left(\sum_{a \in C_{P-1}(\Lambda)} \alpha_a n(i)_a \right) \prod_{a \in C_{P-1}(\Lambda)} V_{\underline{a}(i)}^\perp (n(i)_a) \\ & \quad \times \prod_{b \in C_P(\Lambda)} W_{\underline{b}(i)}^\perp ((dn(i))_b + (-1)^P (s(i+1)_b - s(i)_b) + \delta_{i,0} (\vartheta_\Lambda(\mu^\vee))_b). \end{aligned} \quad (\text{C.42})$$

The decorated Z -kernel is

$$\begin{aligned} & \left\langle s(i) \left| \hat{\mathcal{W}}_\nu^{\delta_{i,0}} \hat{U}_m(\beta^\vee) e^{-\beta \hat{H}_z/M} \right| s(i) \right\rangle \\ &= \chi \left(\int_\nu s(i) \right)^{\delta_{i,0}} \prod_{c \in C_{P+1}(\Lambda)} W_{c(i)}^\parallel ((ds(i))_c + (\vartheta_\Lambda(\beta^\vee))_c) \prod_{b \in C_P(\Lambda)} V_{b(i)}^\parallel (s(i)_b). \end{aligned} \quad (\text{C.43})$$

Define the total electric and magnetic spacetime cycles

$$q_e(\nu, \alpha) := \nu(0) + \text{Sus}(\alpha) \in Z_P(\underline{\Lambda}), \quad (\text{C.44})$$

and

$$q_m(\mu^\vee, \beta^\vee) := \mu^\vee(0) + \text{Sus}(\beta^\vee) \in Z_{d-P}(\underline{\Lambda}^\vee). \quad (\text{C.45})$$

The closure of these cycles follows from $\partial\nu = 0$, $\partial\alpha = 0$, $\partial\mu^\vee = 0$, and $\partial\beta^\vee = 0$, together with periodicity in the Euclidean-time direction.

Combining (C.42) and (C.43), and then using the spacetime packaging (C.25), gives the exact background formula

$$Z_M(\mu^\vee, \nu; \beta^\vee, \alpha) = \sum_{\phi \in \Omega^P(\underline{\Lambda})} \chi \left(\int_{q_e(\nu, \alpha)} \phi \right) \prod_{c \in C_{P+1}(\underline{\Lambda})} W_c((D\phi + \vartheta_{\underline{\Lambda}}(q_m(\mu^\vee, \beta^\vee)))_c) \prod_{u \in C_P(\underline{\Lambda})} V_u(\phi_u). \quad (\text{C.46})$$

This is the desired fused form. The spatial Wilson insertion and the electric Euclidean twist combine into the single spacetime Wilson factor supported on $q_e(\nu, \alpha)$, while the spatial 't Hooft insertion and the magnetic Euclidean twist combine into the single cocycle shift $\vartheta_{\underline{\Lambda}}(q_m(\mu^\vee, \beta^\vee))$. Setting $\mu^\vee = \nu = \beta^\vee = \alpha = 0$ we recover the partition function (C.26) in the trivial background.

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